Membrane Structure
Chapter 10

03.13.26

미생물학교실 박용환

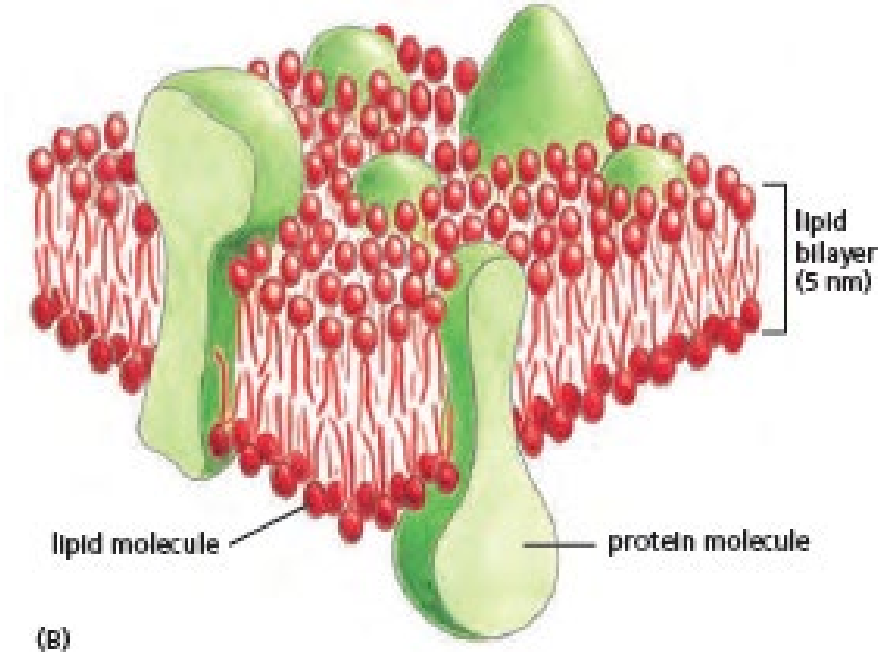
The Lipid Bilayer

❖ 막의 종류,전체 막 면적 대비 비율 (%)
✓ 소포체 (ER, Rough & Smooth) 약 50% ~ 60%
✓ 미토콘드리아 (Mitochondria) 약 20% ~ 30%
✓ 골지체 (Golgi apparatus)약 7%
✓ 세포막 (Plasma membrane),약 2% ~ 5%

Protein : Lipid ≈ 1 : 1 (질량 기준)

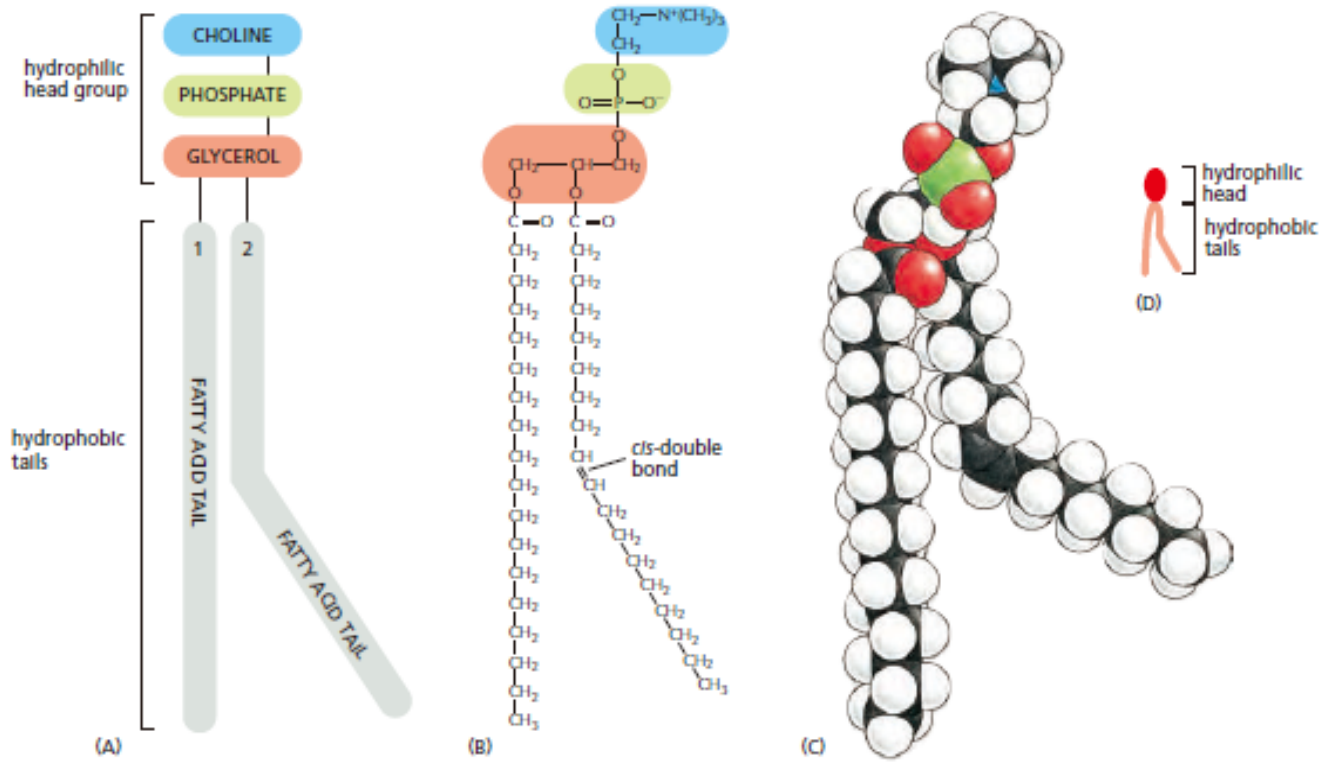
유동성 모자이크 모델, 원형질막, 세포막

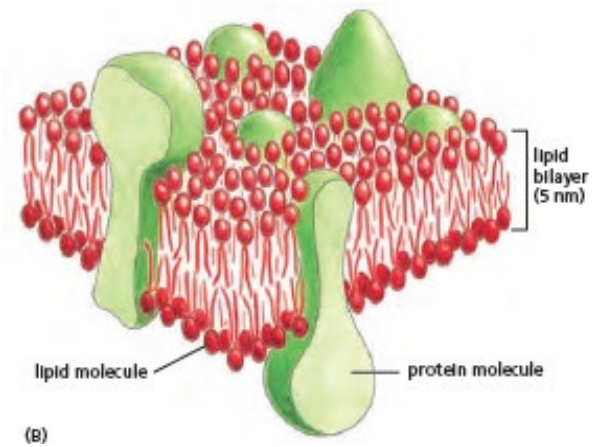
(A) An electron micrograph of a segment of the plasma membrane of a human red blood

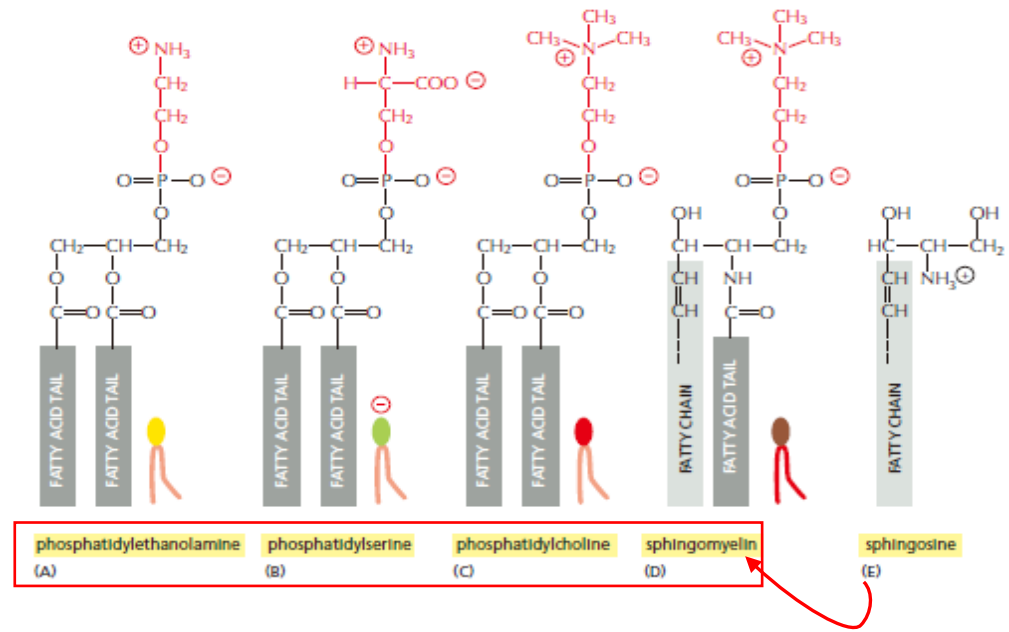


(B) A three-dimensional schematic view of a cell membrane and the general disposition of its lipid and protein constituents.

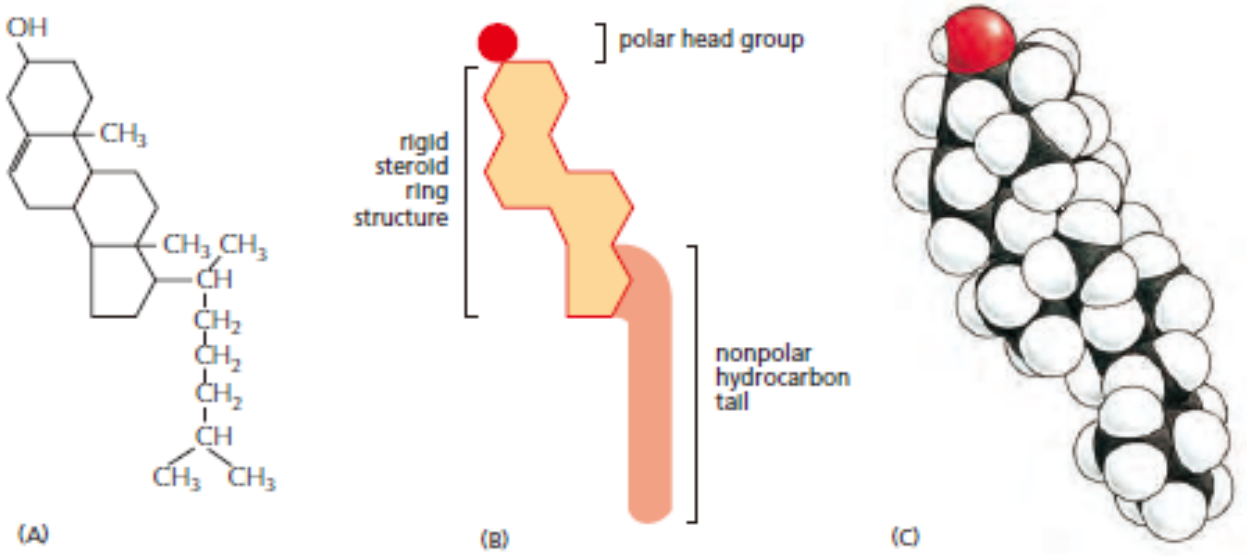
Phosphoglycerides, Sphingolipids, and Sterols Are the Major Lipids in Cell Membranes

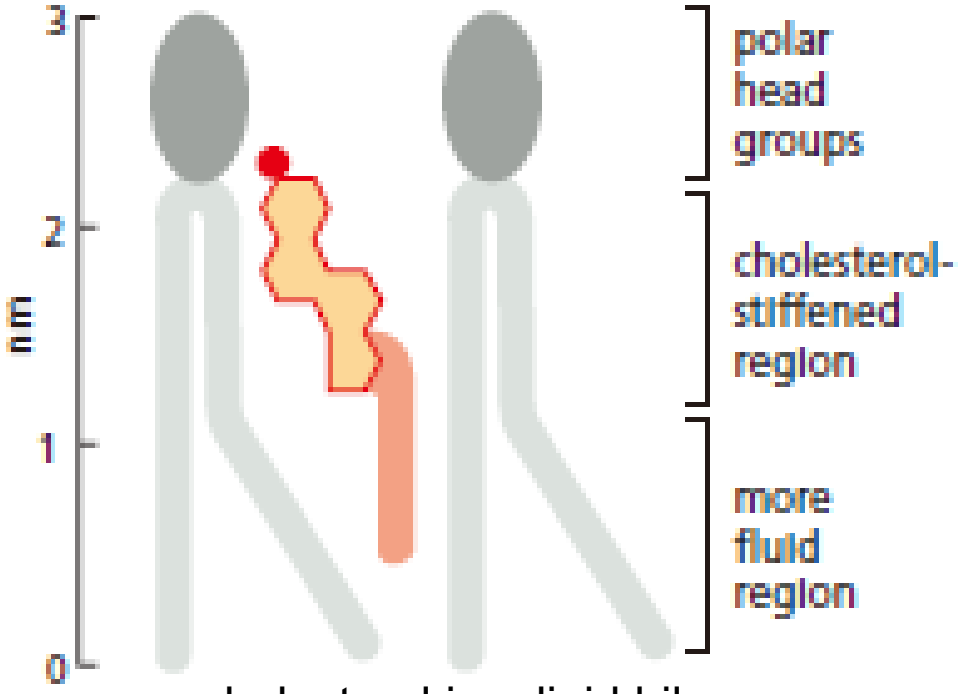






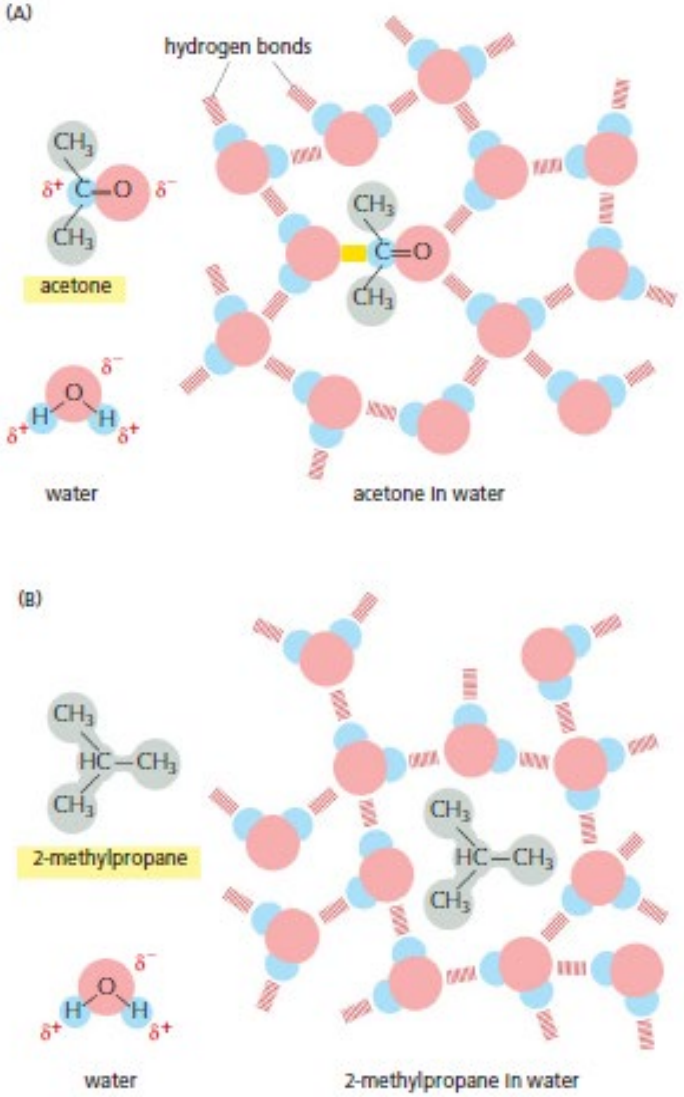
The structure of cholesterol

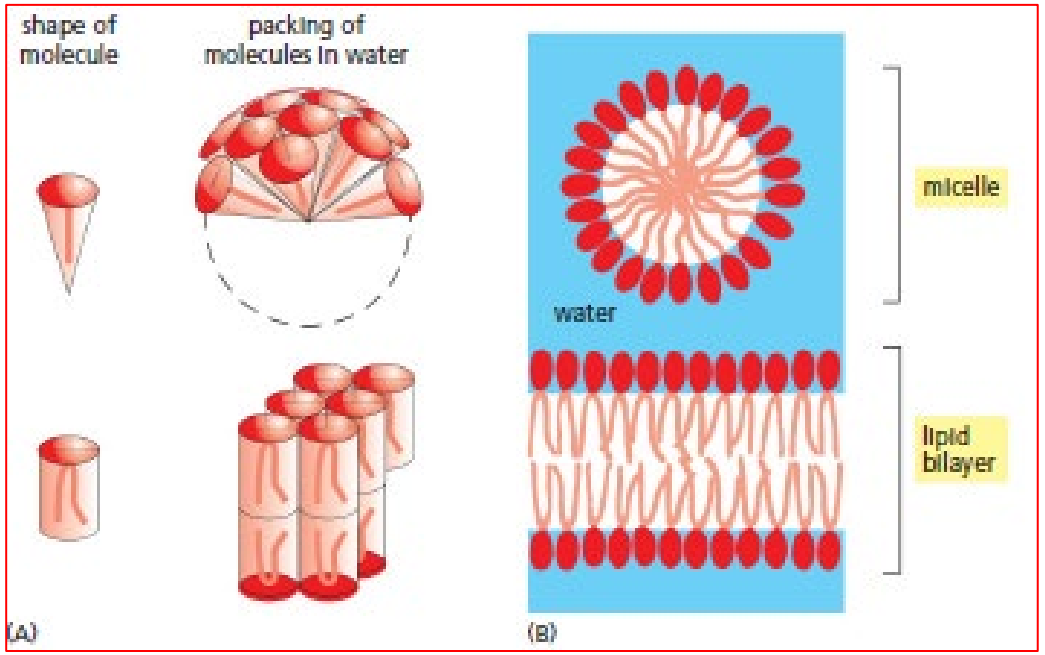


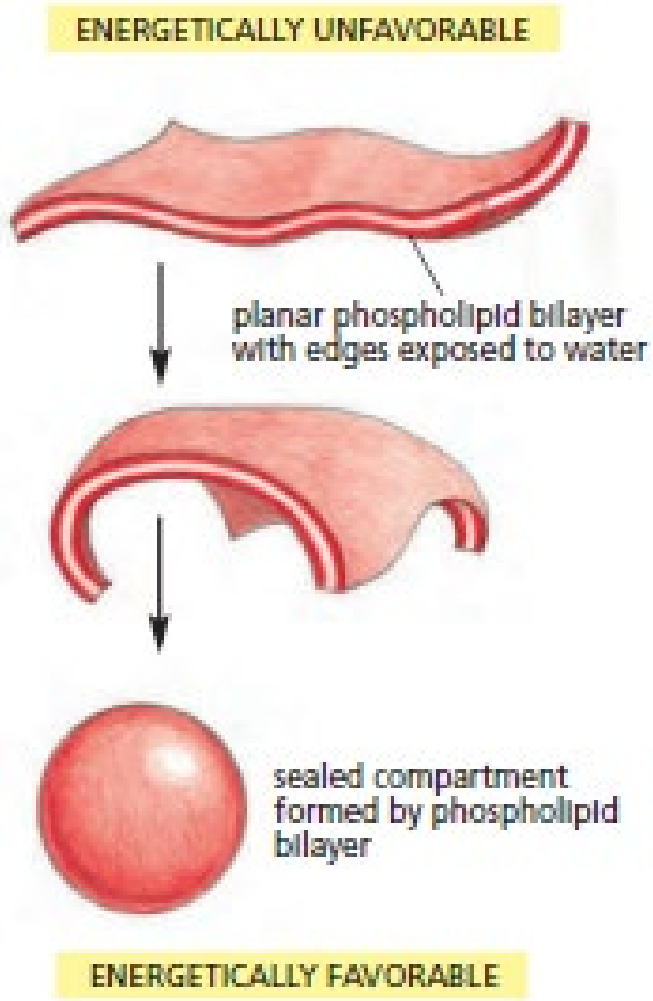


cholesterol in a lipid bilayer

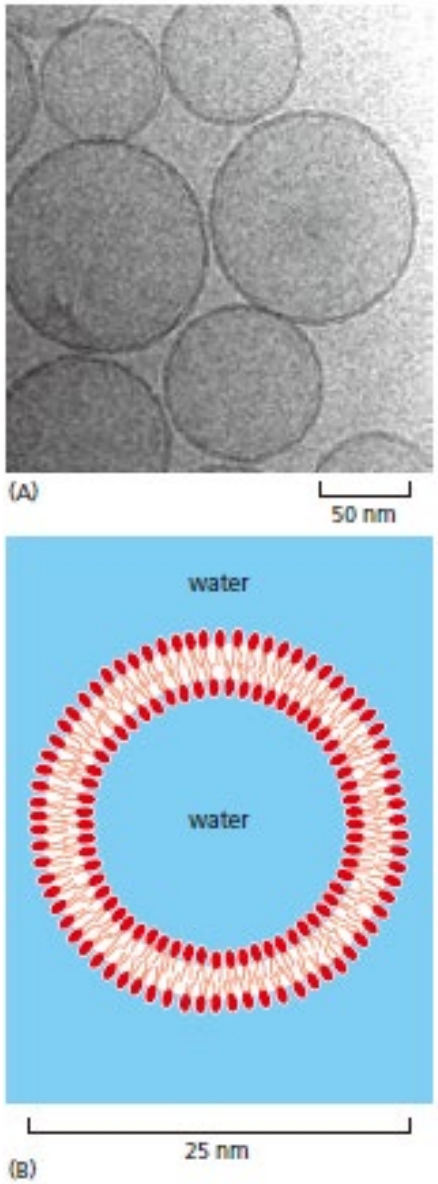
How hydrophilic and hydrophobic molecules interact differently with water?





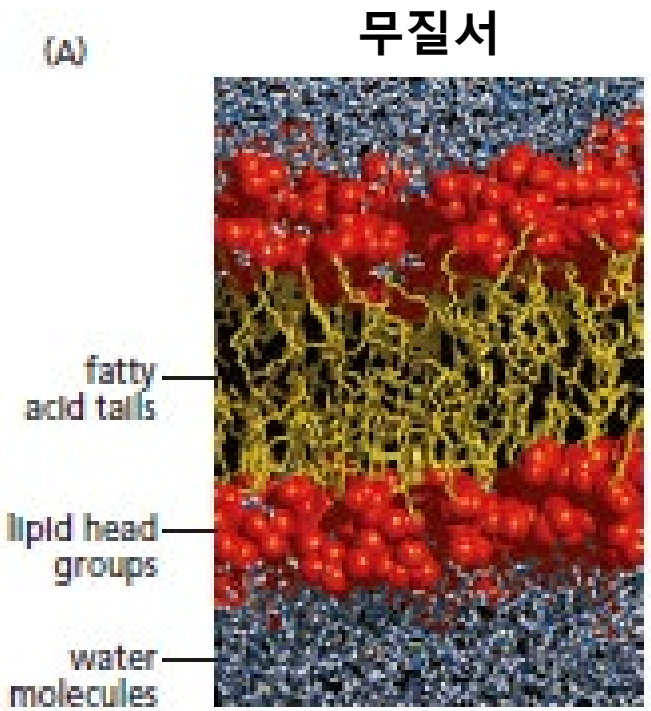


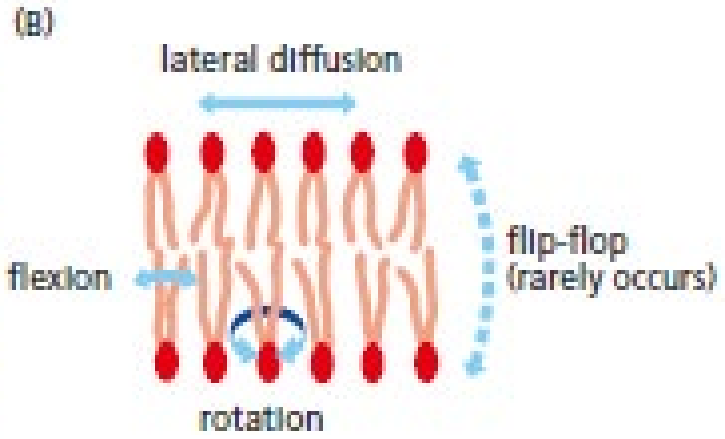
The Lipid Bilayer Is a Two-dimensional Fluid



합성 인지질
liposome in D.W

무질서





1) lateral diffusion(평형 확산)
2) Rotation
3) Flexion (꼬리 부분이 유연하게 움직이는 것)
4) flip-flop (거의 일어나지 않음)

The Fluidity of a Lipid Bilayer Depends on Its Composition

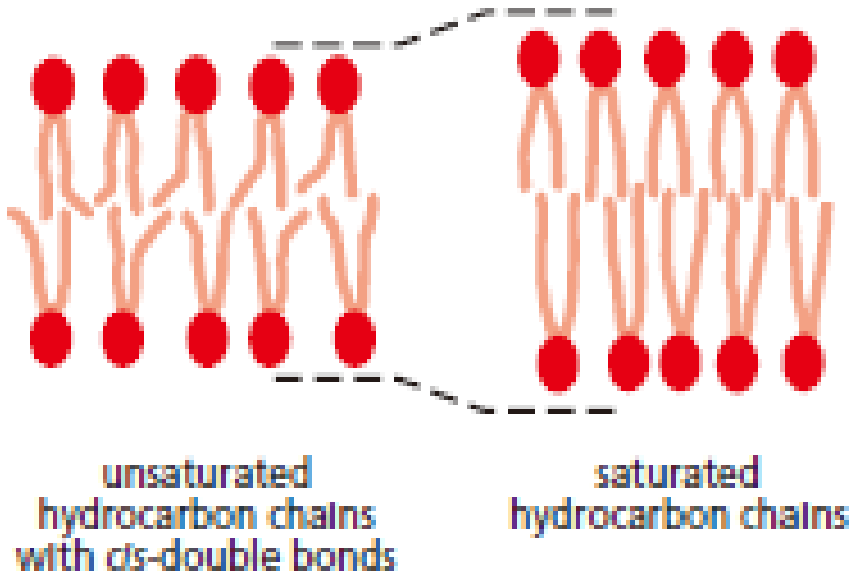


TABLE 10–1 Approximate Lipid Compositions of Different Cell Membranes

Lipid	Percentage of total lipid by weight					
	Liver cell plasma membrane	Red blood cell plasma membrane	Myelin	Mitochondrion (inner and outer membranes)	Endoplasmic reticulum	*E. coli* bacterium
Cholesterol	17	23	22	3	6	0
Phosphatidylethanolamine	7	18	15	28	17	70
Phosphatidylserine	4	7	9	2	5	trace
Phosphatidylcholine	24	17	10	44	40	0
Sphingomyelin	19	18	8	0	5	0
Glycolipids	7	3	28	trace	trace	0
Others	22	14	8	23	27	30

✓ 막 단백질의 수송 기능, 효소 활성막 단백질 이동, vesicle 형성 및 막 융합

✓ 단세포 생물 – 온도 변화에 따라 지방산 조성 바꿈

이중층의 유동성: 구성& 온도

❖ 세포막의 유동성에 영향을 미치는 요소는 세가지

✓ **지방산의 포화 정도: 불포화 지방산 → 이중결합 → 강한 결합 방해**

✓ **온도**

✓ **지방산의 길이:** hydrophobic interaction의 증가 → 단단한 결합, 유동성 약화

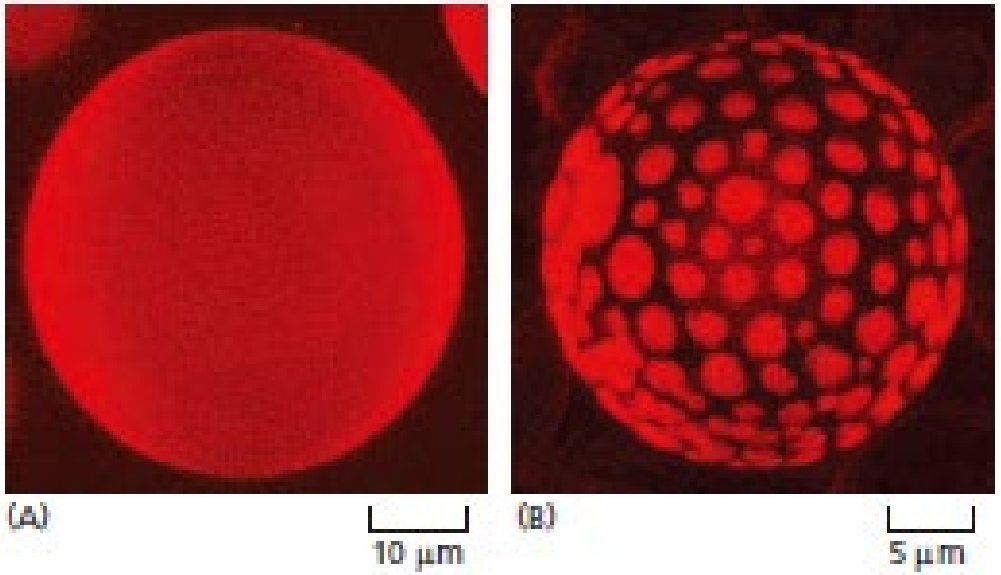
❖ 콜레스테롤 역할

✓ 지질 사이 빈틈 차단 – 작은 분자 투과성 감소

✓ **온도가 낮을 때: 막의 경화 차단**

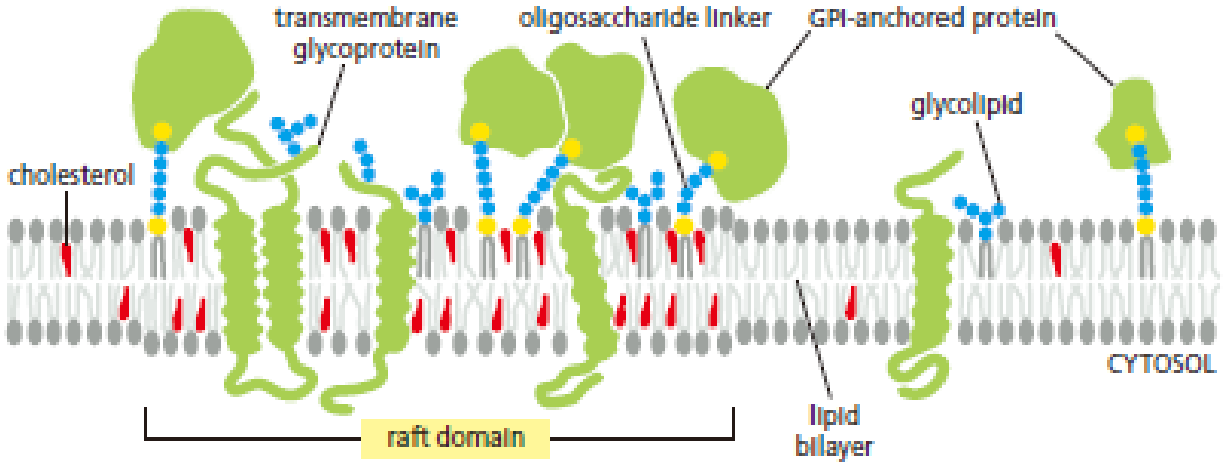
✓ **온도가 높을 때: 막의 유동성 차단**

Despite Their Fluidity, Lipid Bilayers Can Form Domains of Different Compositions



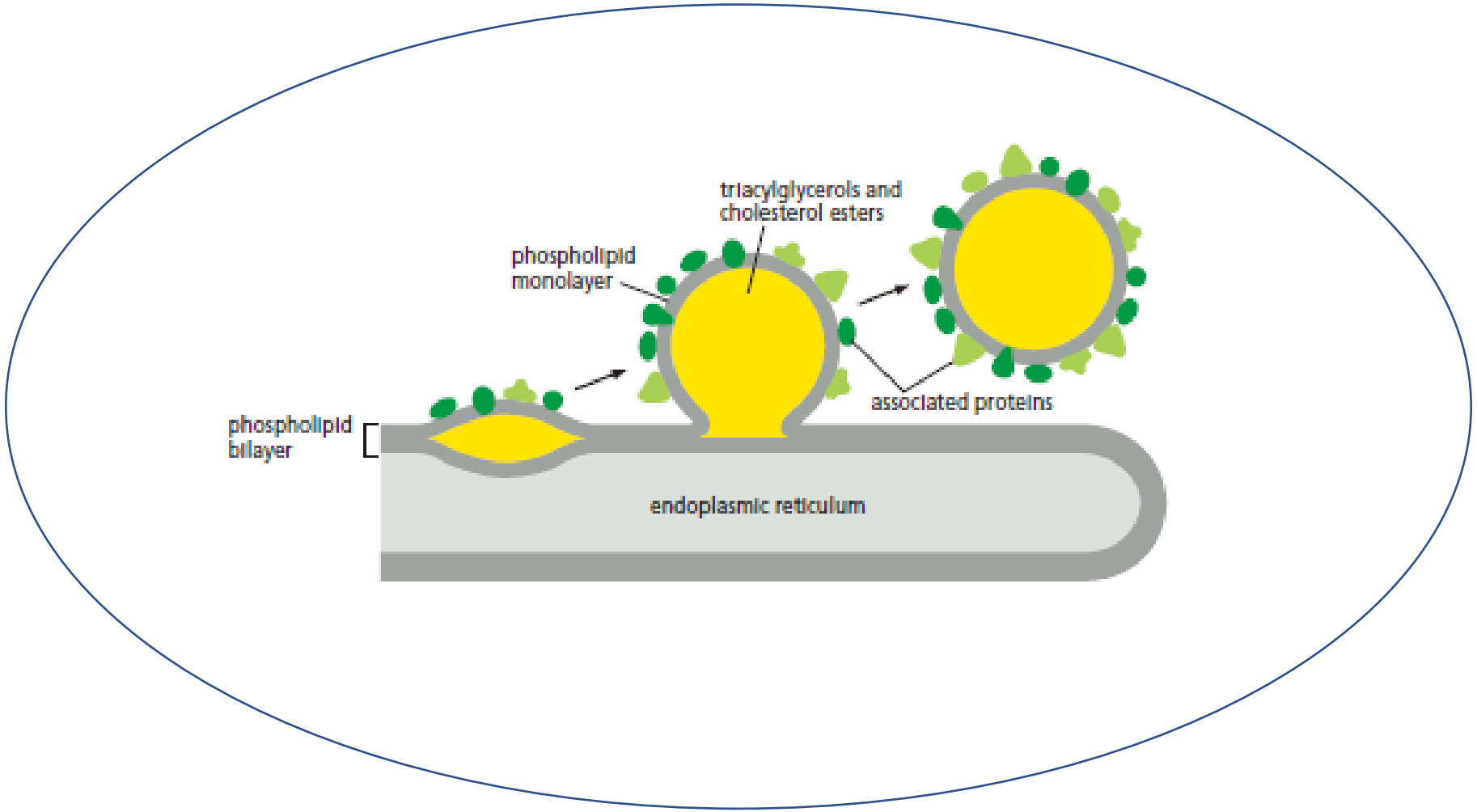
Uniform bilayers
1:1 mixture of
phosphatidylcholine
and sphingomyelin

Two separate phases
1:1:1 mixture of
phosphatidylcholine,
sphingomyelin,
and cholesterol

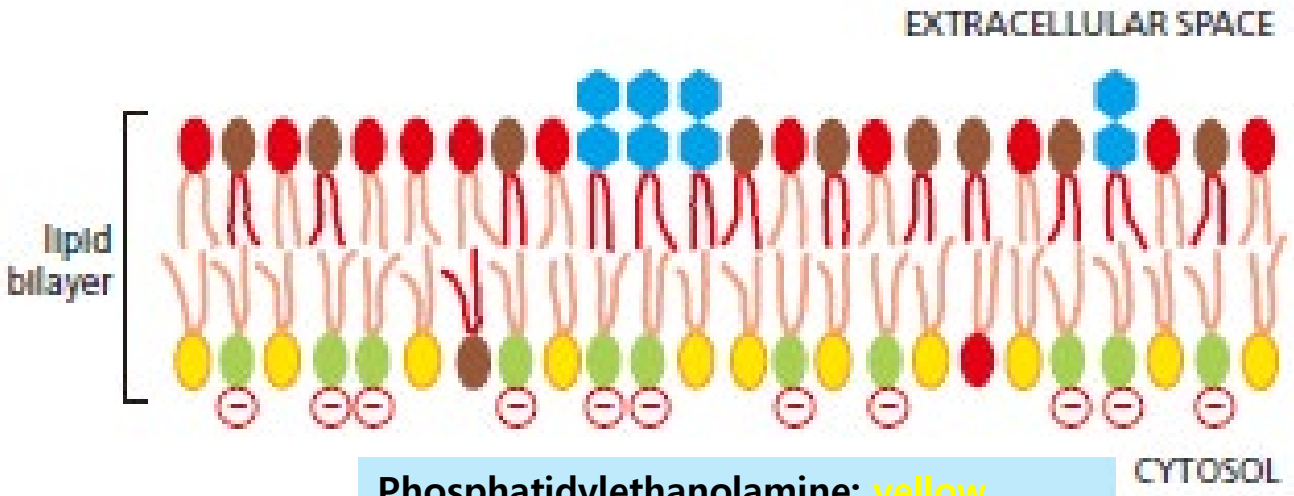


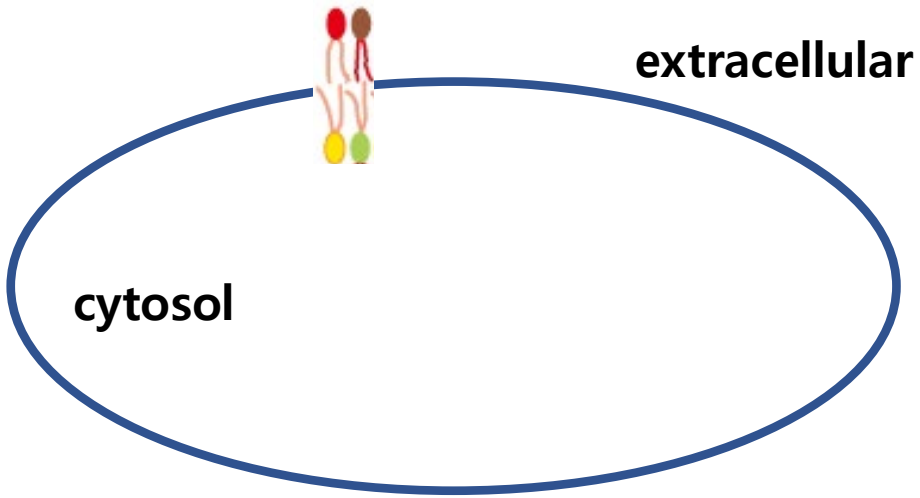
- ✓ 실제 세포막은 동적인 시스템
  이라 상 분리가 유지되지 않음
- ✓ Lipid raft

Lipid Droplets Are Surrounded by a Phospholipid Monolayer



The Asymmetry of the Lipid Bilayer Is Functionally Important





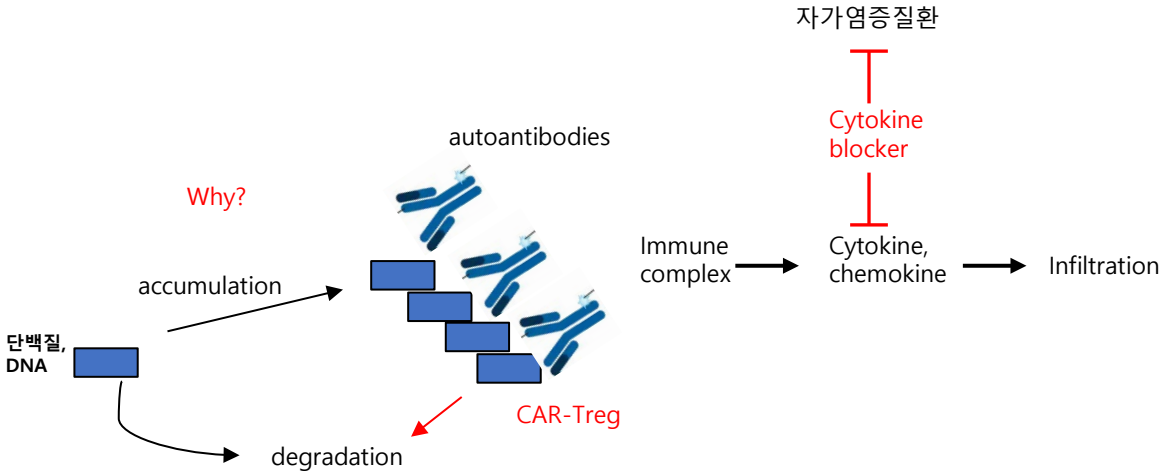
Phosphatidylethanolamine: yellow
Phosphatidylserine: green
Phosphatidylcholine: red
Sphingomyelin: brown
Glycolipid: blue

apoptotic 세포에서 phosphatidylserine의 이동 메커니즘

1. 이 지질을 외부 단층에서 내부 단층으로 정상적으로 수송하는 인지질 이동이 비활성화

2. 인지질을 두 단층 사이에서 양방향으로 비특이적으로 전달하는 "scramblase"가 활성화

---> phosphatidylserine 만 계속적으로 외부로 노출



형성평가

❖ 이 표를 보고 판단할 수 있는 것은?

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형성평가

Lipid	*E. coli* bacterium
Cholesterol	0
Phosphatidylethanolamine	70
Phosphatidylserine	trace
Phosphatidylcholine	0
Sphingomyelin	0
Glycolipids	0
Others	30

- 최근 새로 발견된 Gasdermin D (GSDMD)라는 단백질은 세포질 영역에 존재하고 있다가 특정 효소에 의해 절단된 후 세포막의 음극성 인지질에 결합해 membrane에 구멍을 만든다. 이 단백질 (GSDMD)이 세포밖으로 나왔을 때 할 수 있는 역할로 타당한 것은?

① 세포막에 박혀 경화를 억제시킨다.

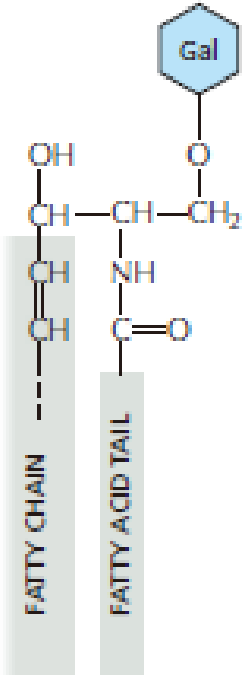
② 박테리아 세포막의 콜레스테롤에 결합한다.

③ 이웃한 세포로 이동하여 세포막에 구멍을 만든다.

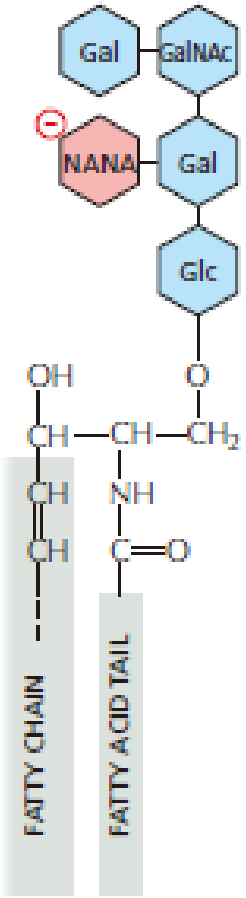
④ 침투해온 박테리아의 세포막 외부에 직접 결합하여 구멍을 만든다.

⑤ 침투한 박테리아의 내부로 들어가 음극성 인지질에 결합해 구멍을 만든다.

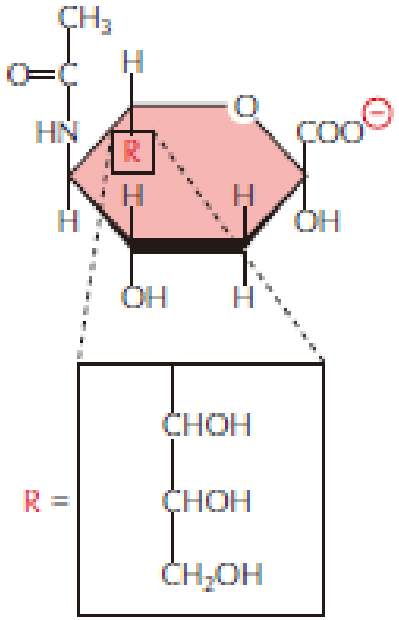
Glycolipids Are Found on the Surface of All Eukaryotic Plasma Membranes



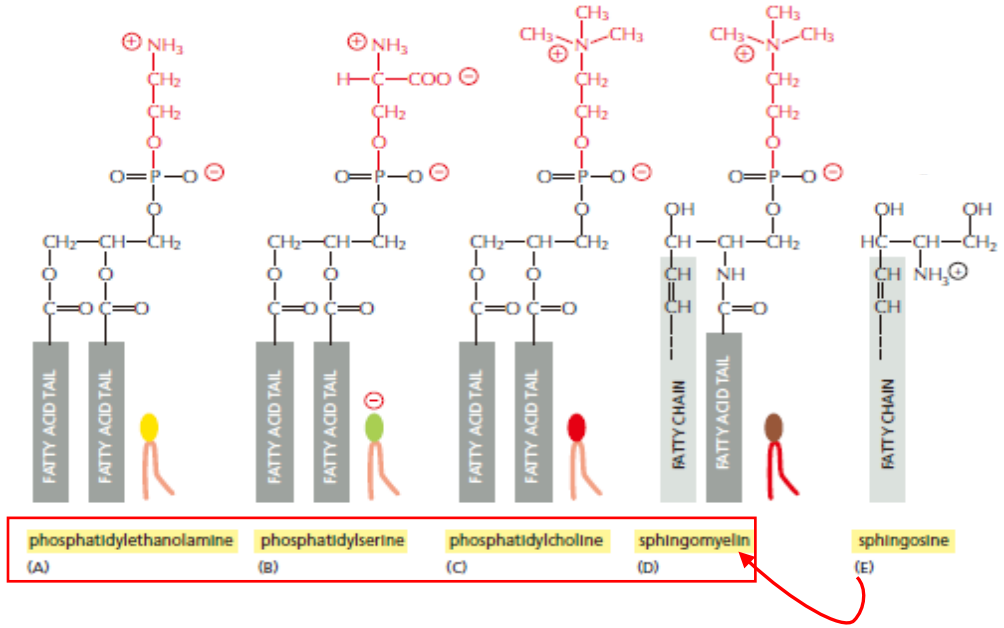
(A) galactocerebroside



(B) G_{M1} ganglioside

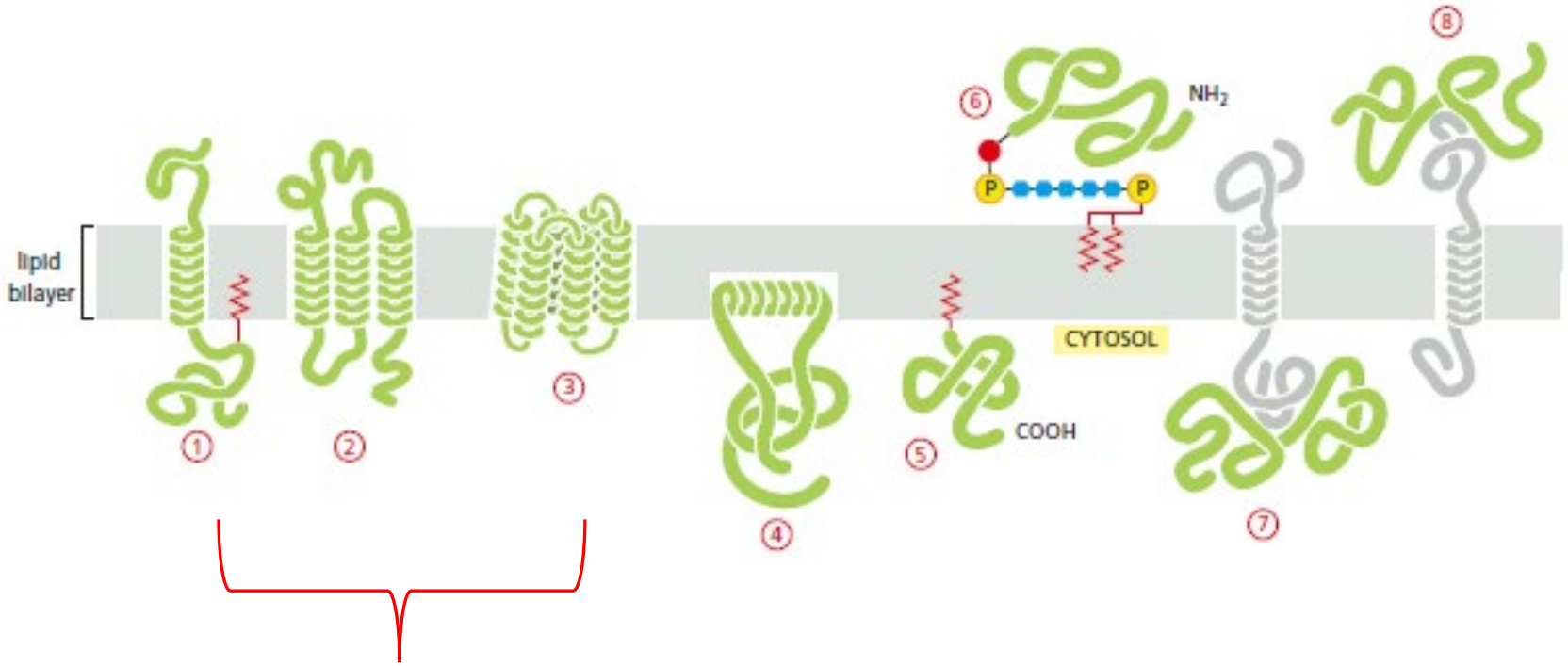


(C) a sialic acid (NANA)

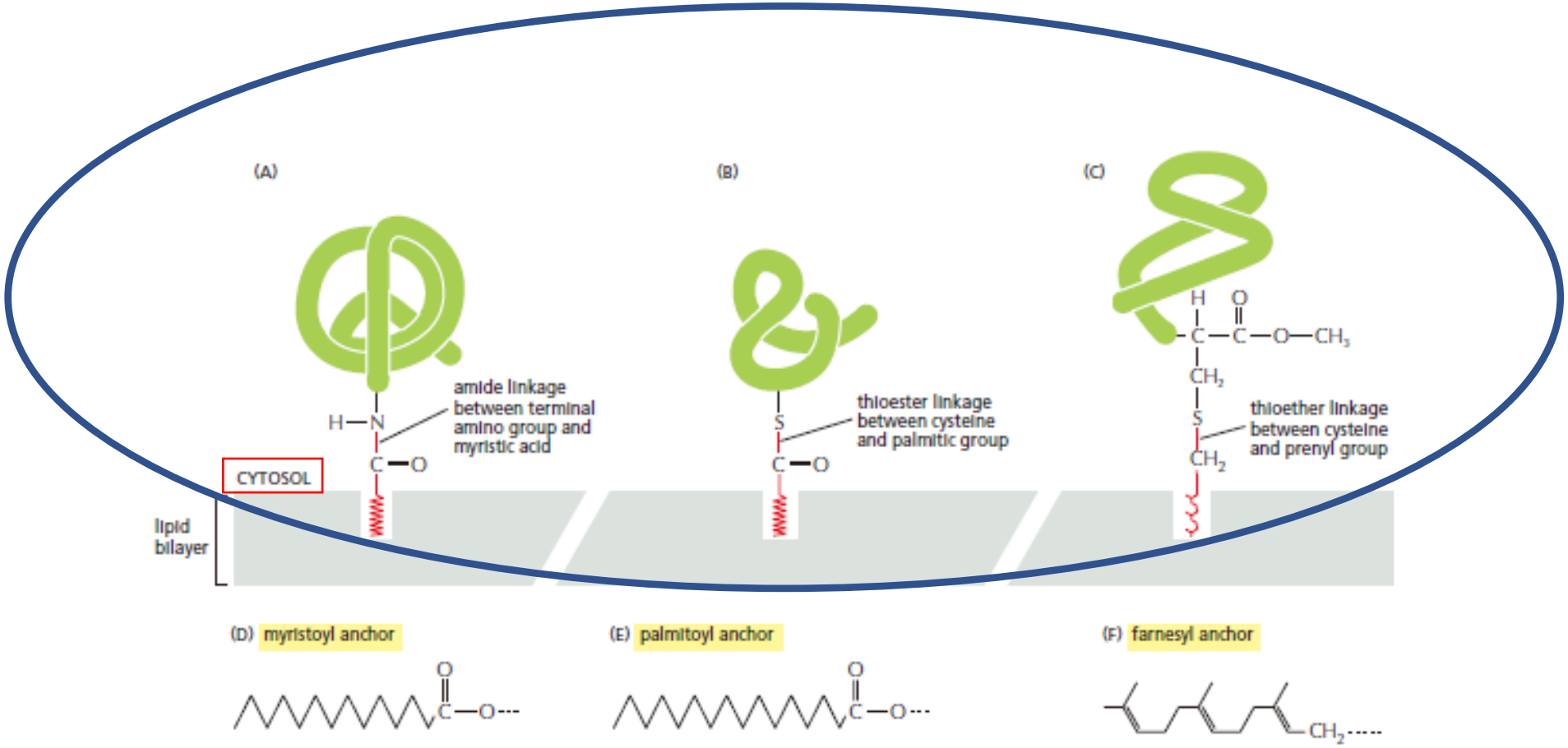


Membrane Proteins

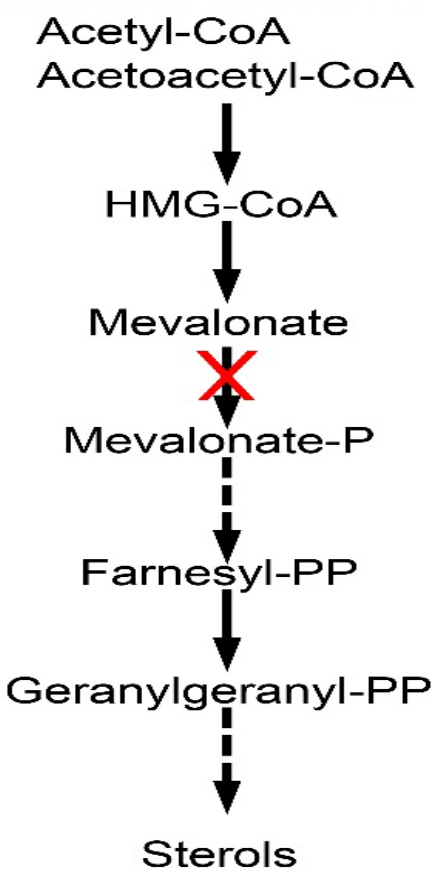
Membrane Proteins Can Be Associated with the Lipid Bilayer in Various Ways



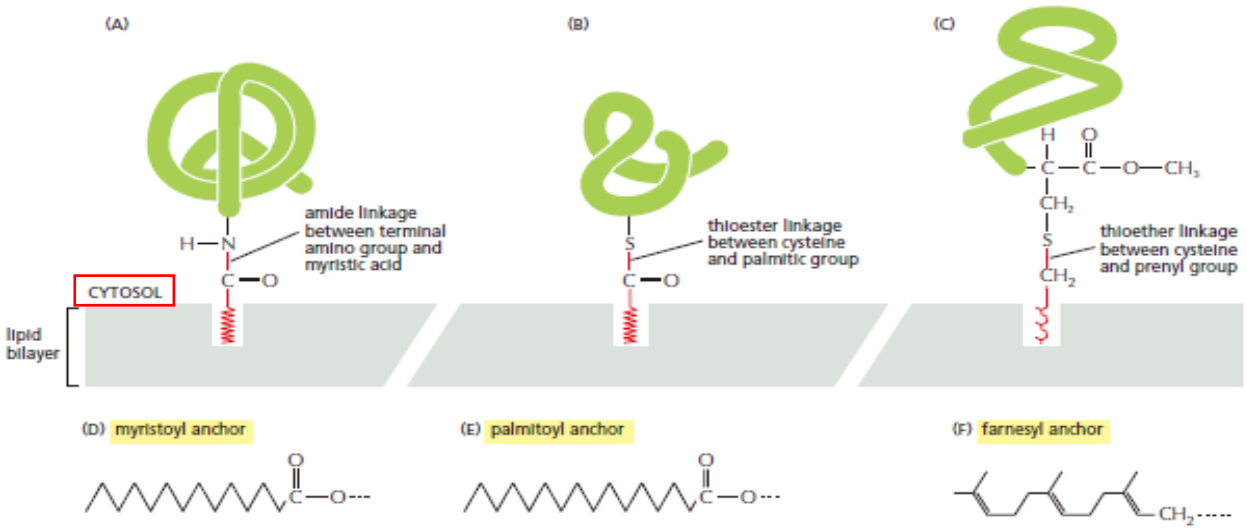
Lipid Anchors Control the Membrane Localization of Some Signaling Proteins



형성평가

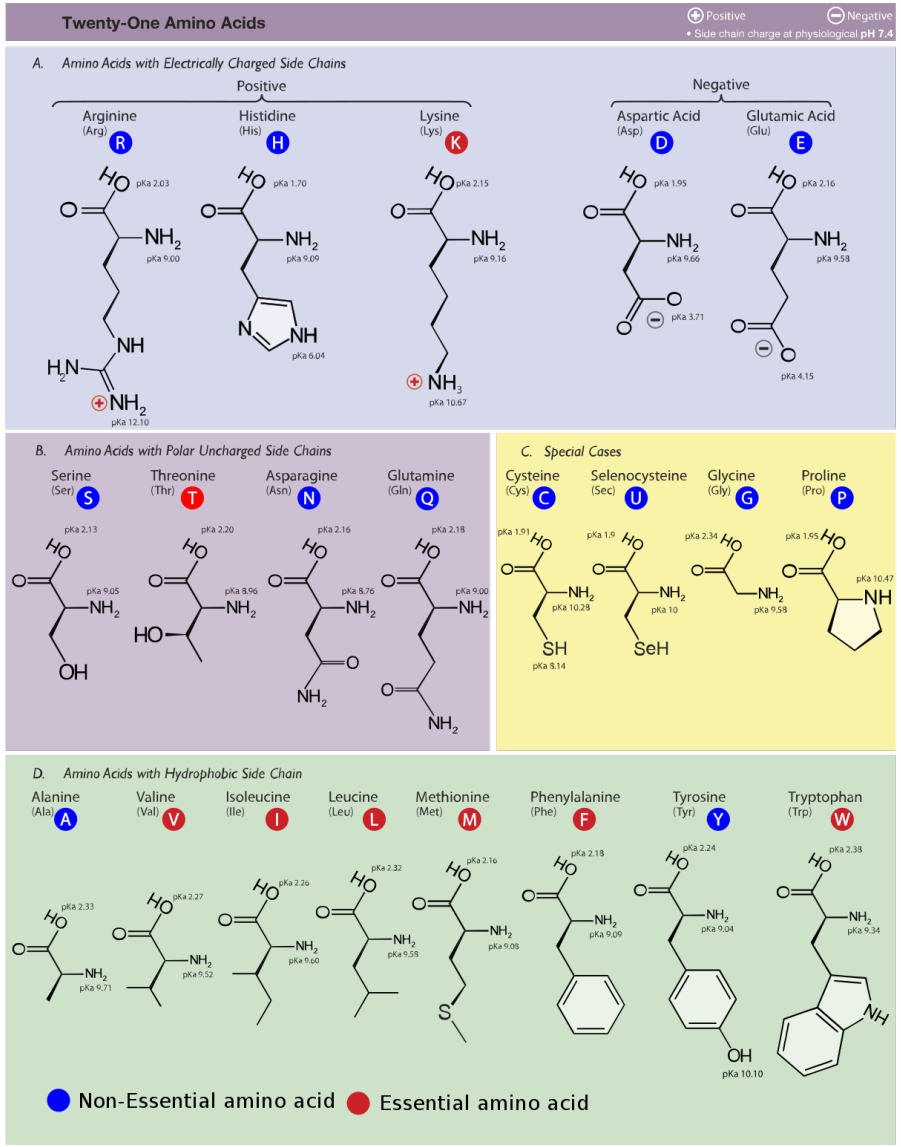


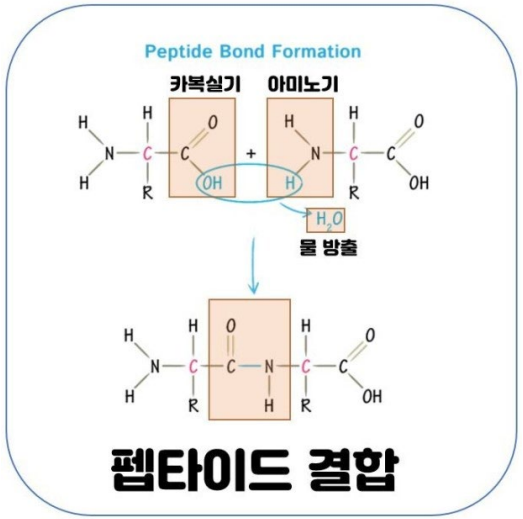
희귀 질환 중에 mevalonate kinase deficiency 라는 질환이 있다. 이 질환의 환자들은 특정 단백질이 세포막에 고정되지 못한다는 사실이 밝혀졌다. 이유를 추론하라.

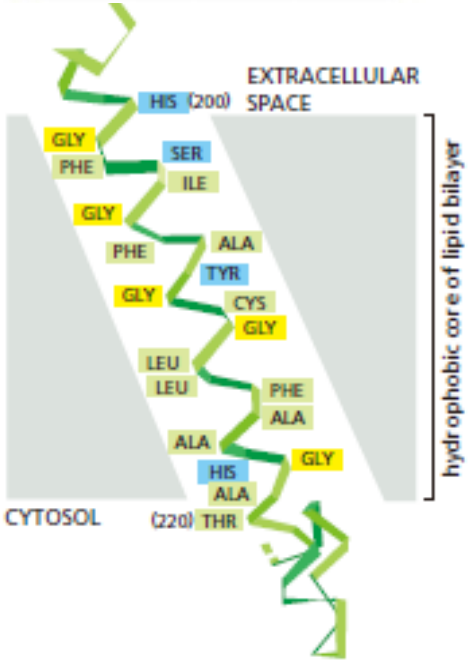


In Most Transmembrane Proteins, the Polypeptide Chain Crosses the Lipid Bilayer in an α-Helical Conformation

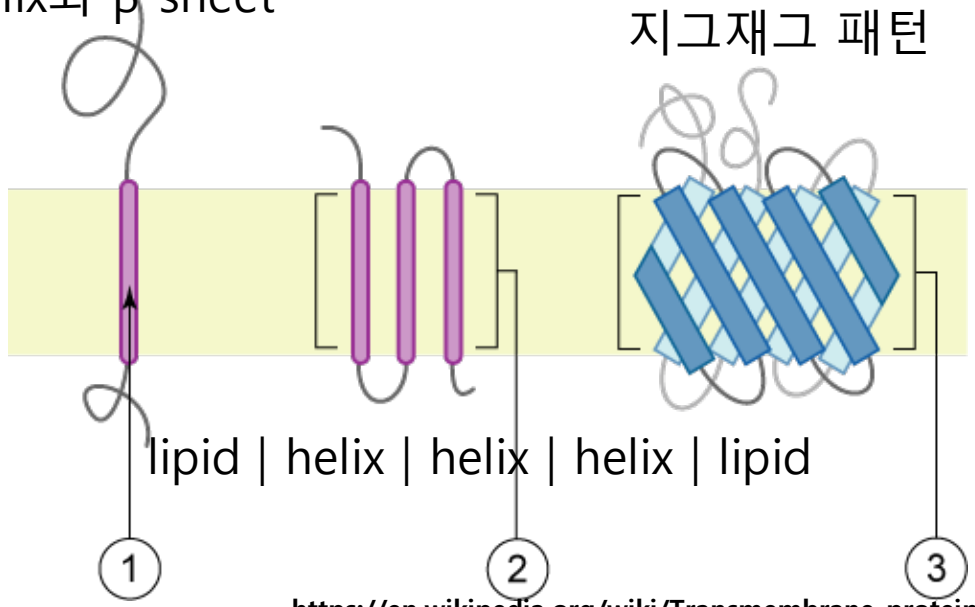
막 통과 단백질의 특별한 구조: α-helix와 β-sheet







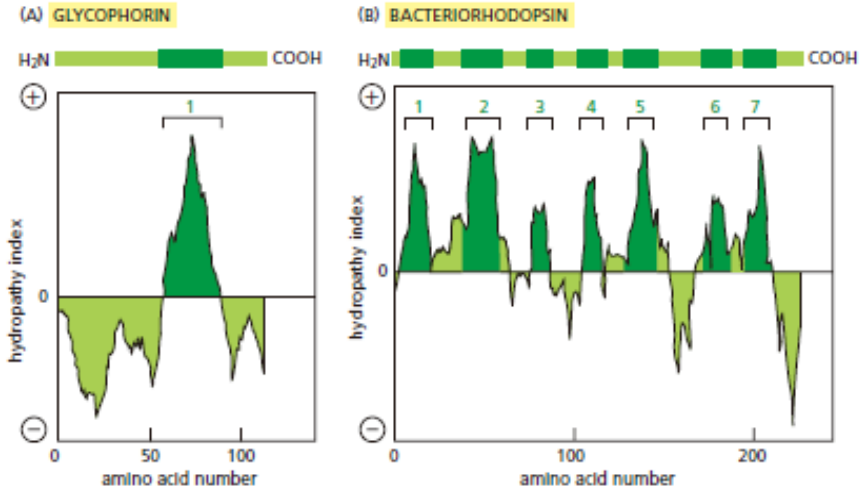
지그재그 패턴



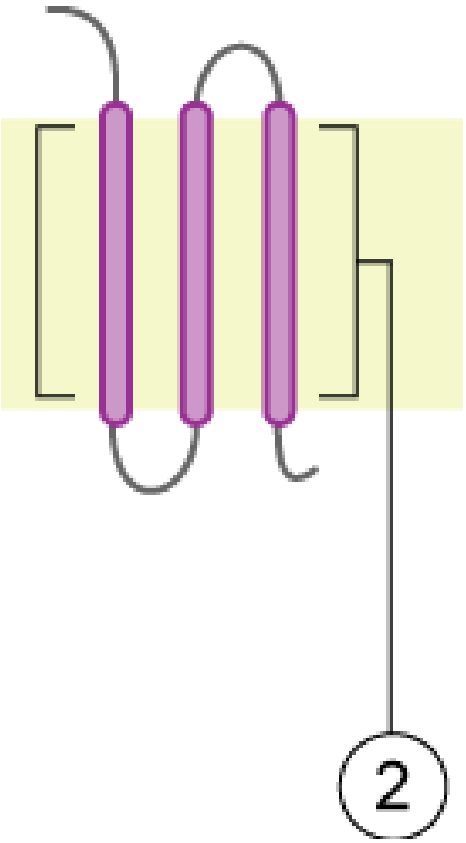
https://en.wikipedia.org/wiki/Transmembrane_protein

positive value: free energy is required for transfer to water

Hydropathy plots



Structural Characteristics of Pore-Forming α-Helical Conformation



❖ 알파 헬릭스는 한 바퀴(360°) 완전히 회전하는 데 약 3.6개의 아미노산이 필요.

대략 아미노산 1개 = 100°

Pore 형성을 위한 헬릭스는 한쪽 면은 친수성, 반대쪽 면 (lipid와 접촉하는 쪽)은 소수성

Ex>

✓ **1번 아미노산:** 구멍 안쪽(친수성)

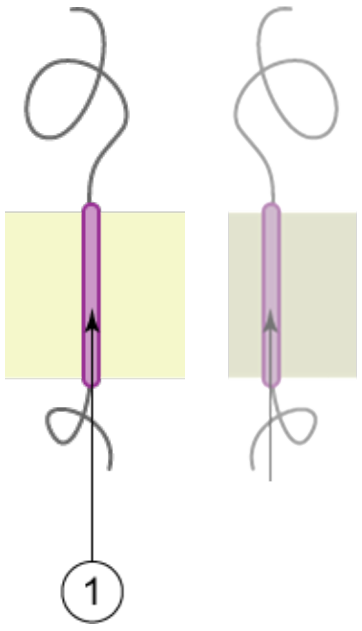
✓ **2번 아미노산:** 100° 회전 (여전히 안쪽 또는 측면)

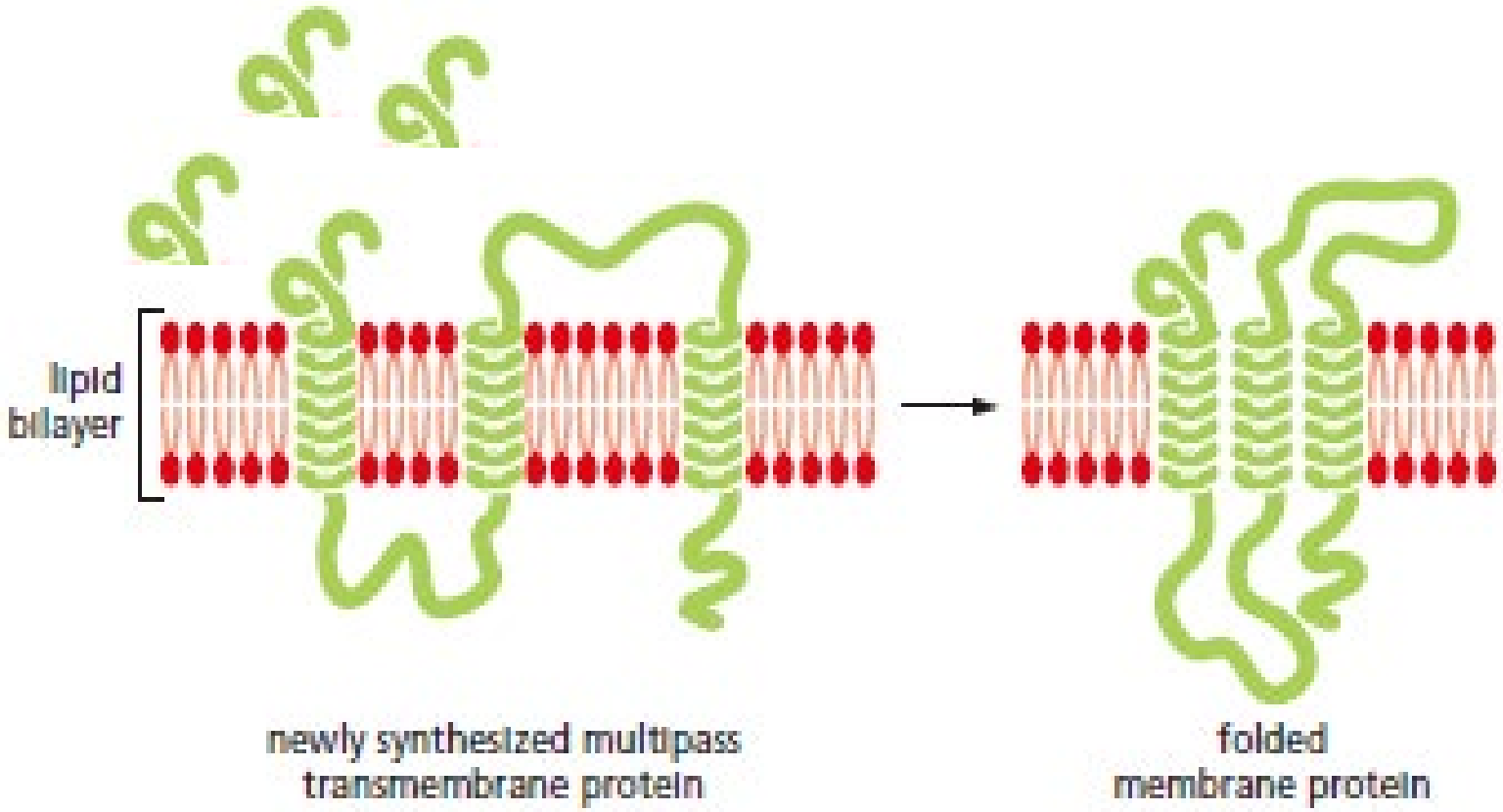
✓ **3번 아미노산:** 또 100° 회전 (이제 바깥쪽 지질을 향하기 시작함 - 소수성)

✓ **4번 아미노산:** 또 100° 회전 (완전히 바깥쪽 지질 쪽 - 소수성)

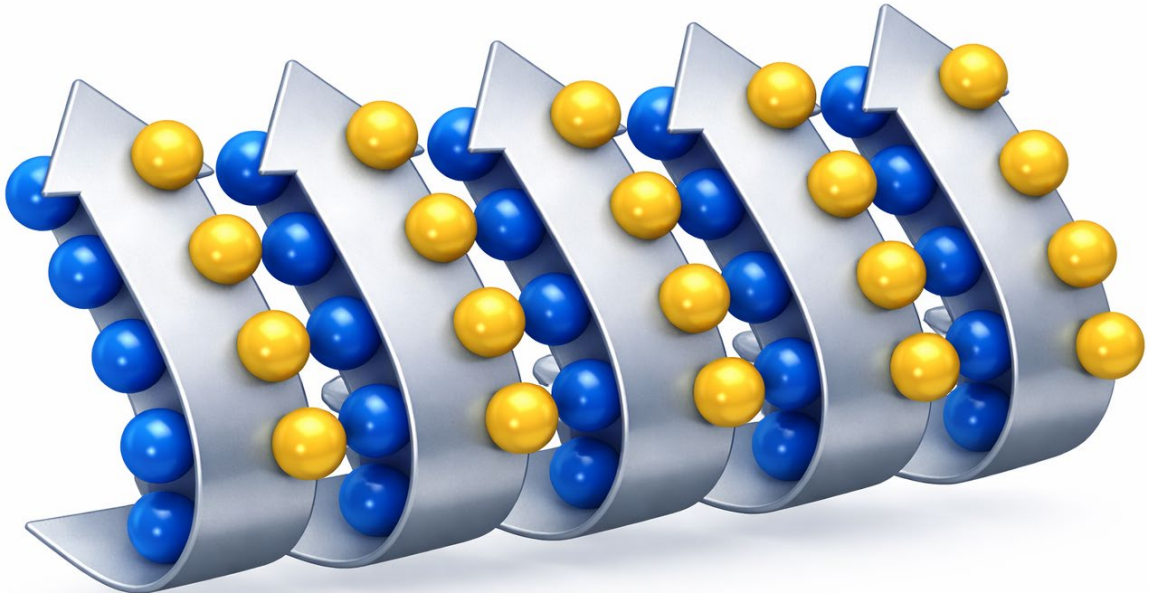
→ **3~4개를 주기로 친수성, 소수성이 교대로 나타나면**, 헬릭스가 계속 돌아가더라도 **특정 각도 범위(예: 0°~180°)에는 친수성 아미노산만, 나머지 각도(180°~360°)에는 소수성 아미노산만** 모일 수 있음.

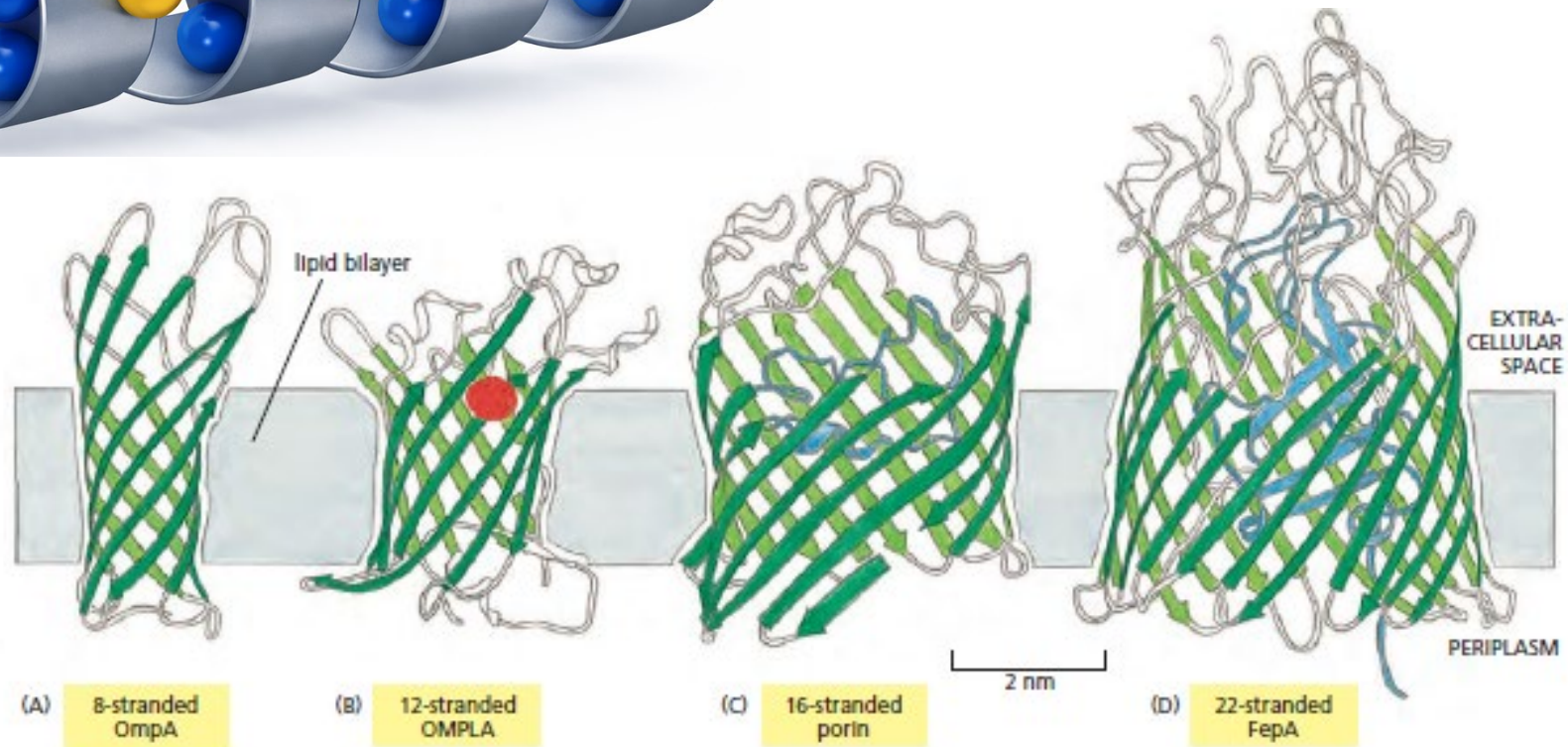
Transmembrane α Helices Often Interact with One Another





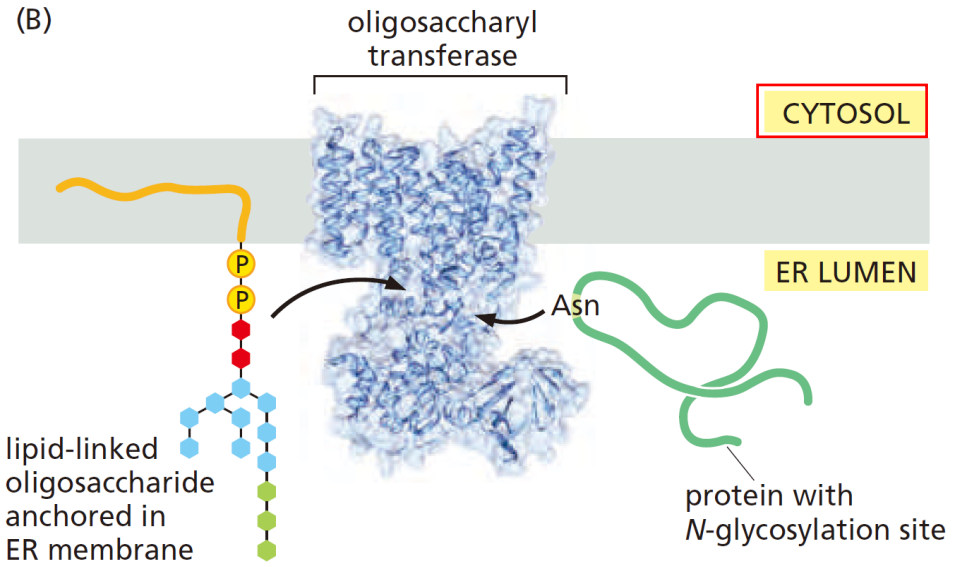
Some β Barrels Form Large Channels

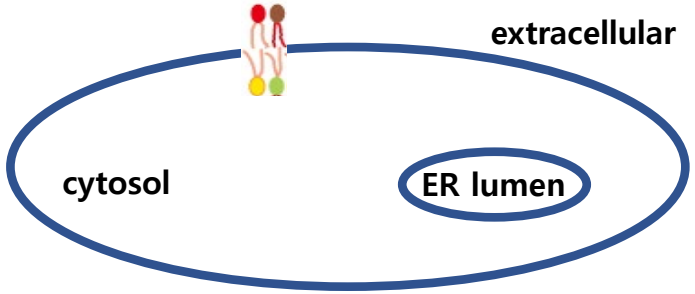


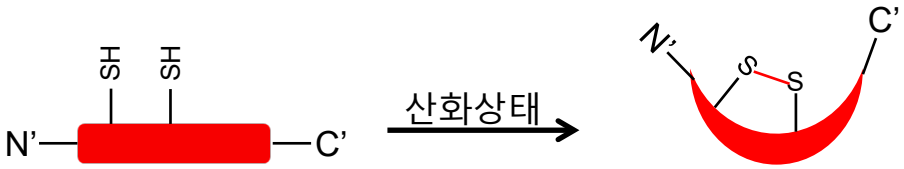


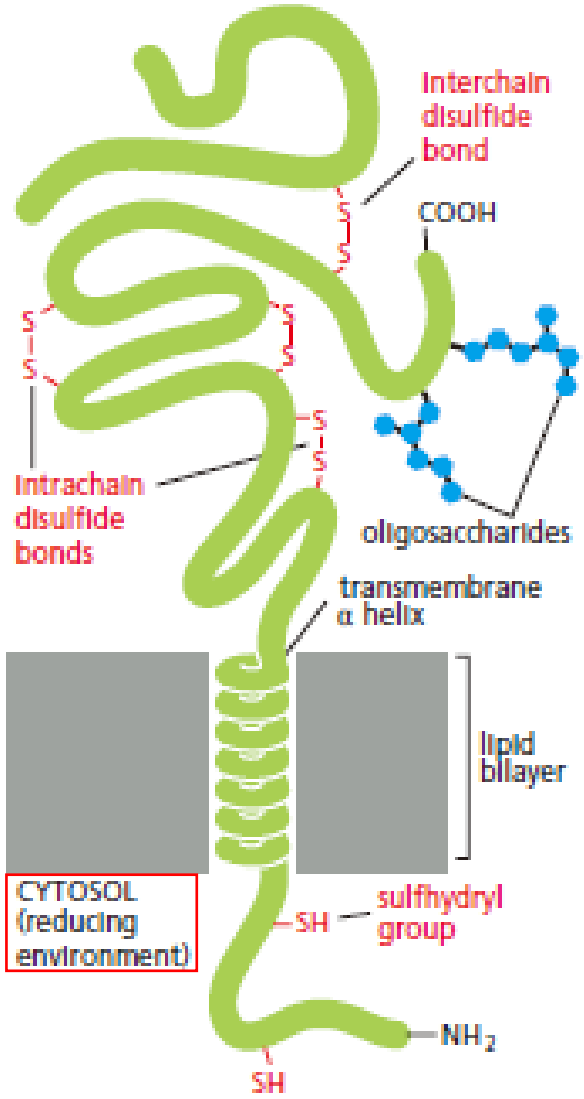
Many Membrane Proteins Are Glycosylated

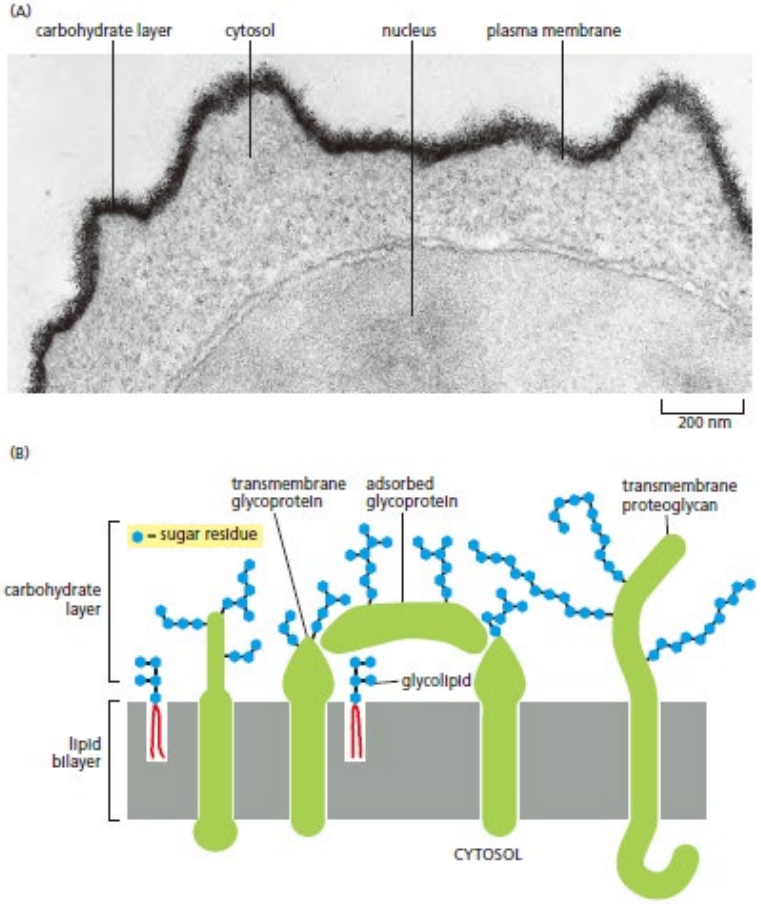
Lipid에 있는 당을 protein으로 옮겨주는 역할



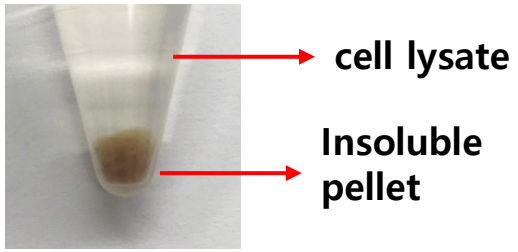


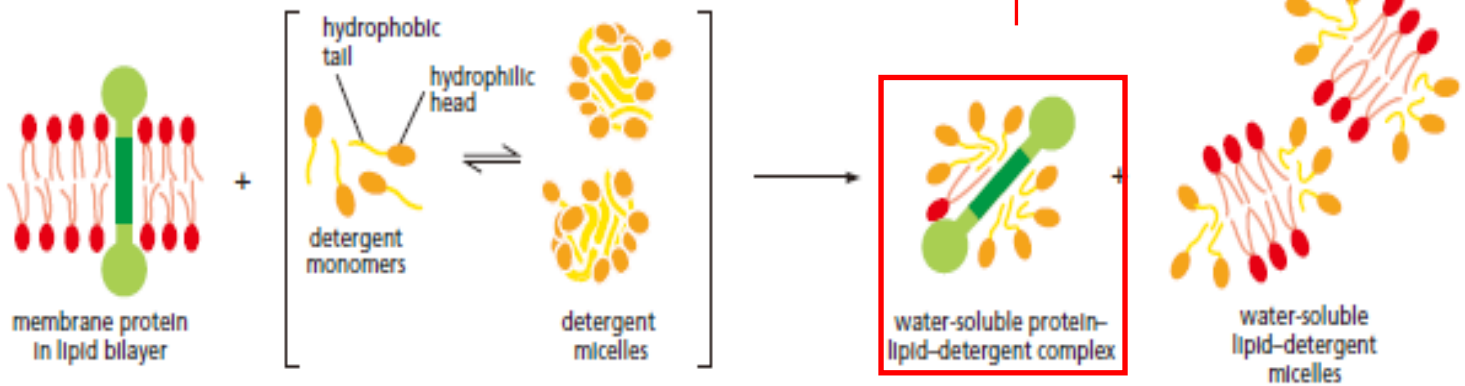


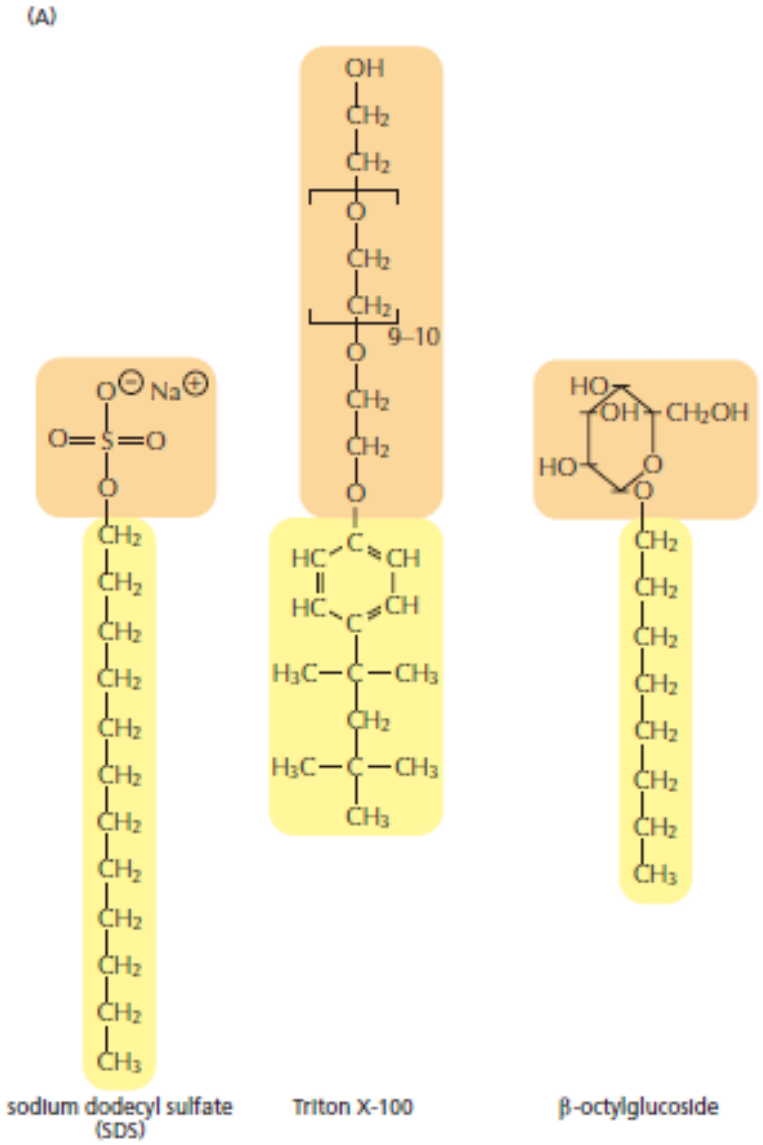




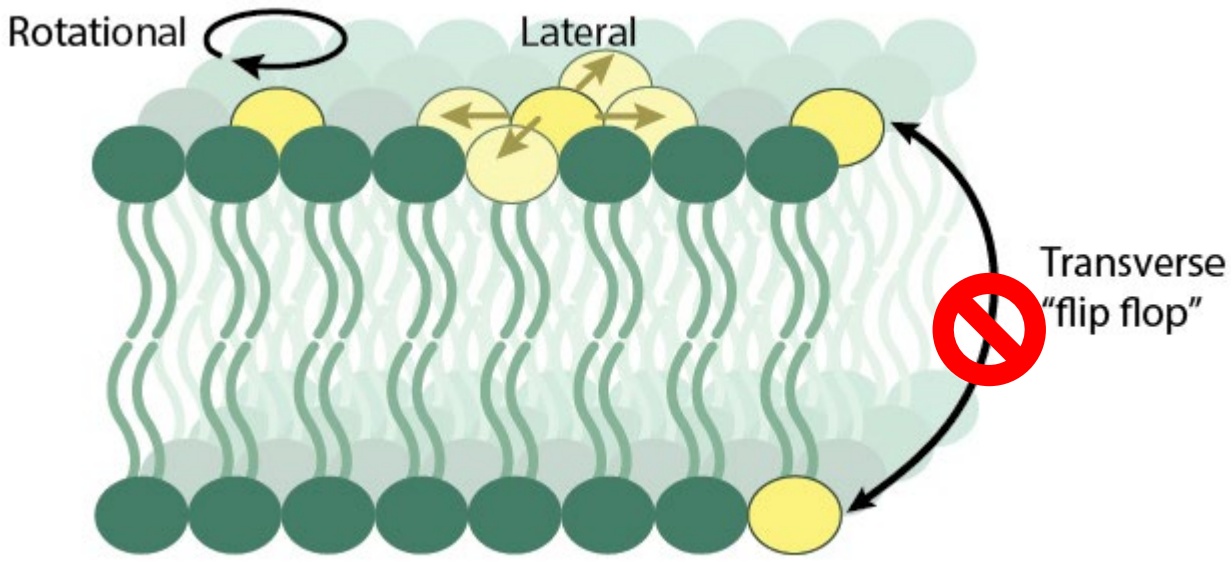
Membrane Proteins Can Be Solubilized and Purified in Detergents

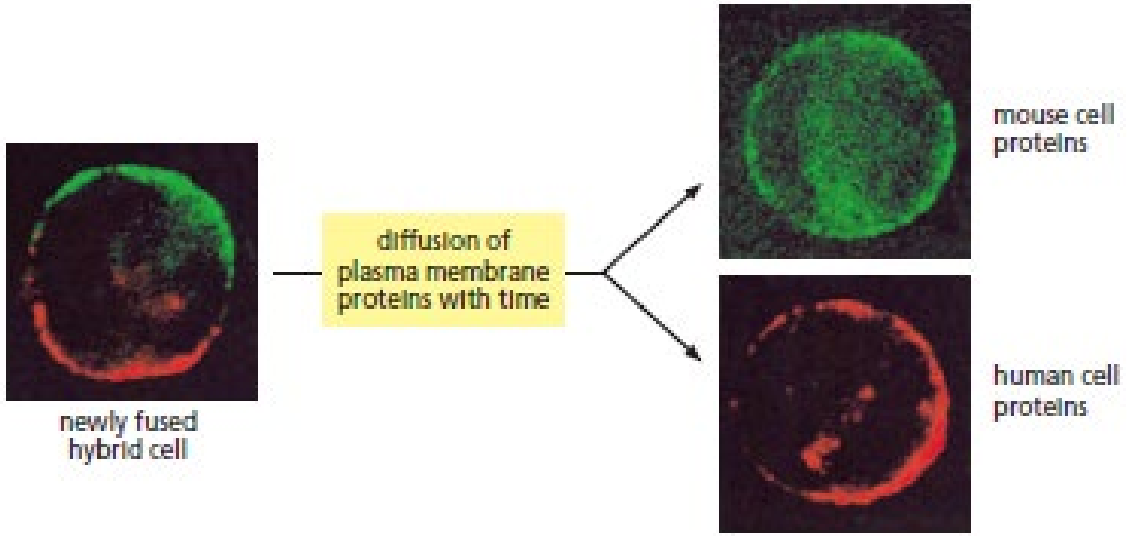






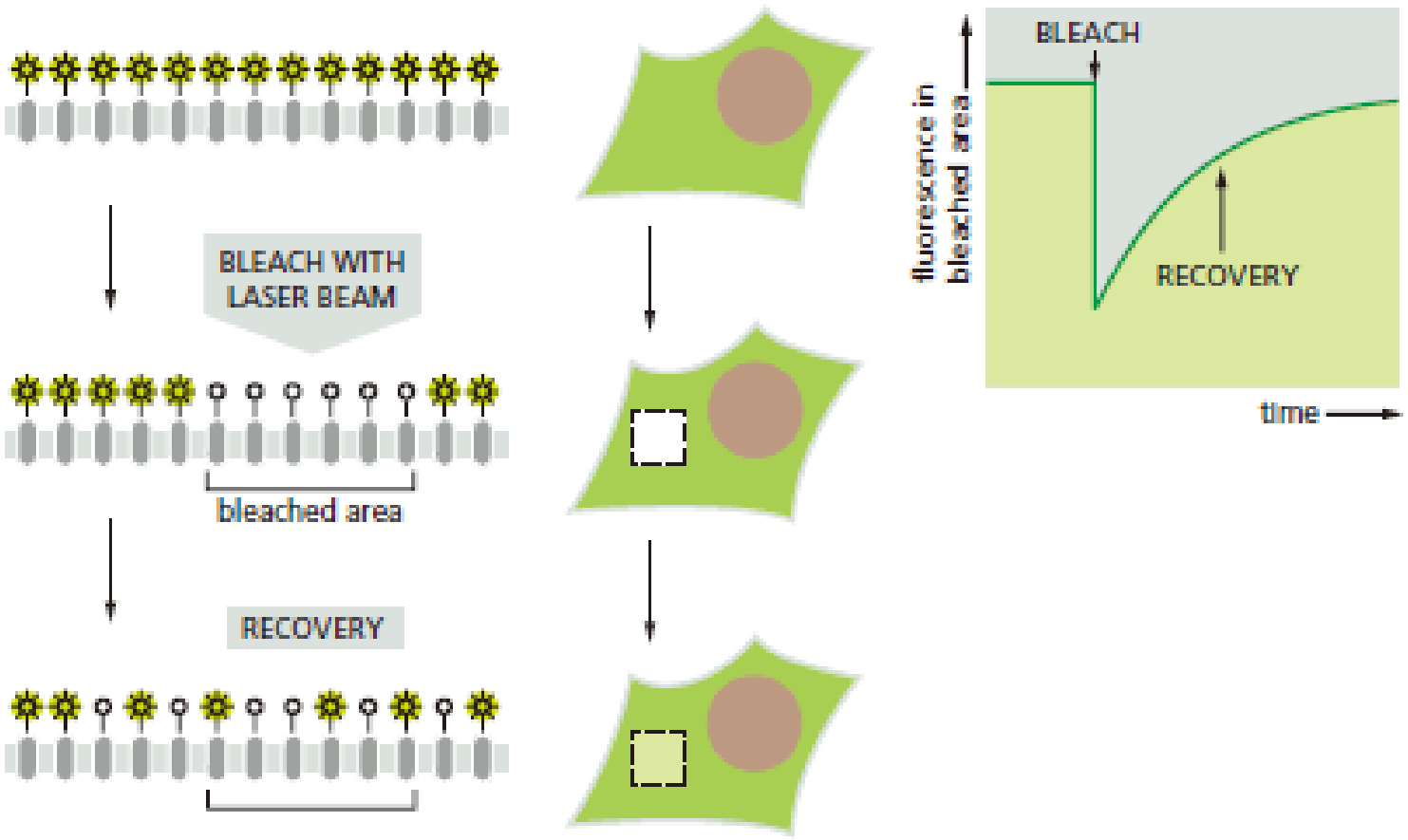
Many Membrane Proteins Diffuse in the Plane of the Membrane



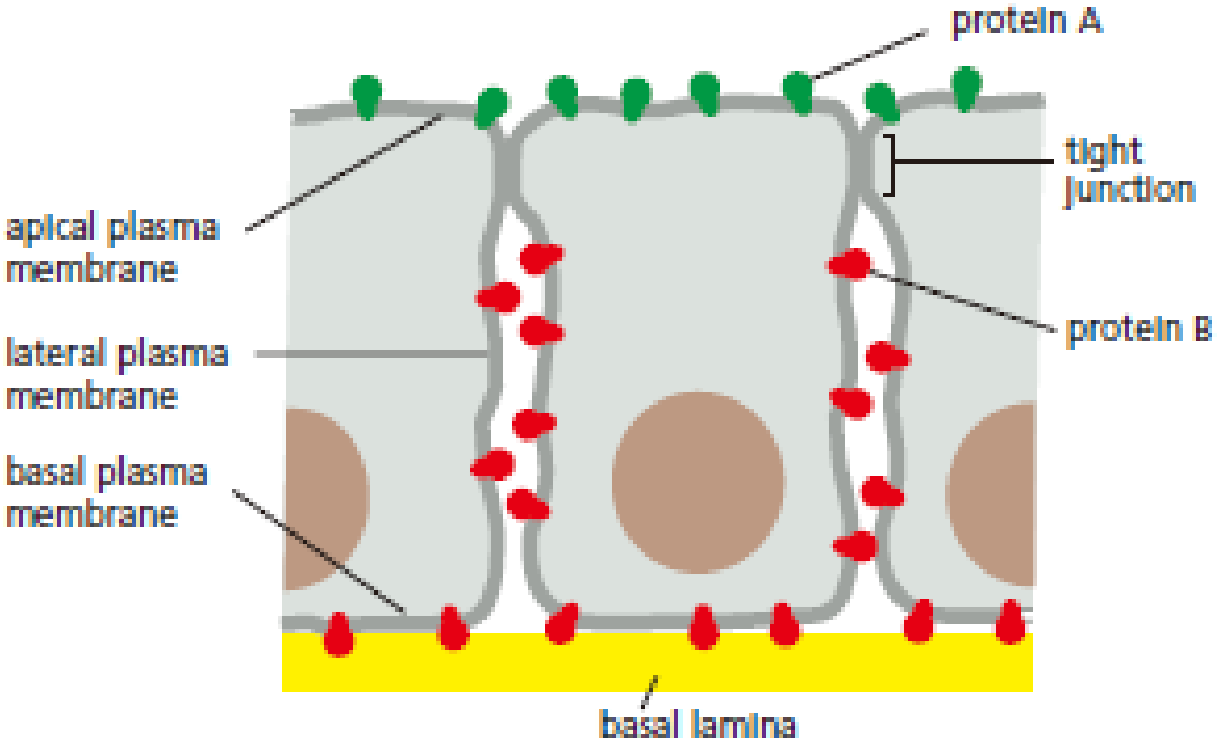


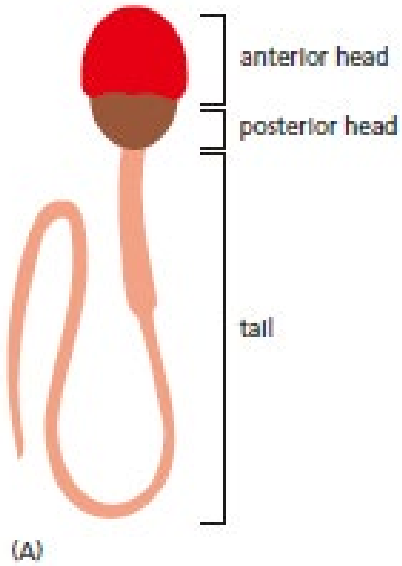
Green **dye: mouse plasma membrane proteins,**
Red **dye: human plasma membrane proteins**

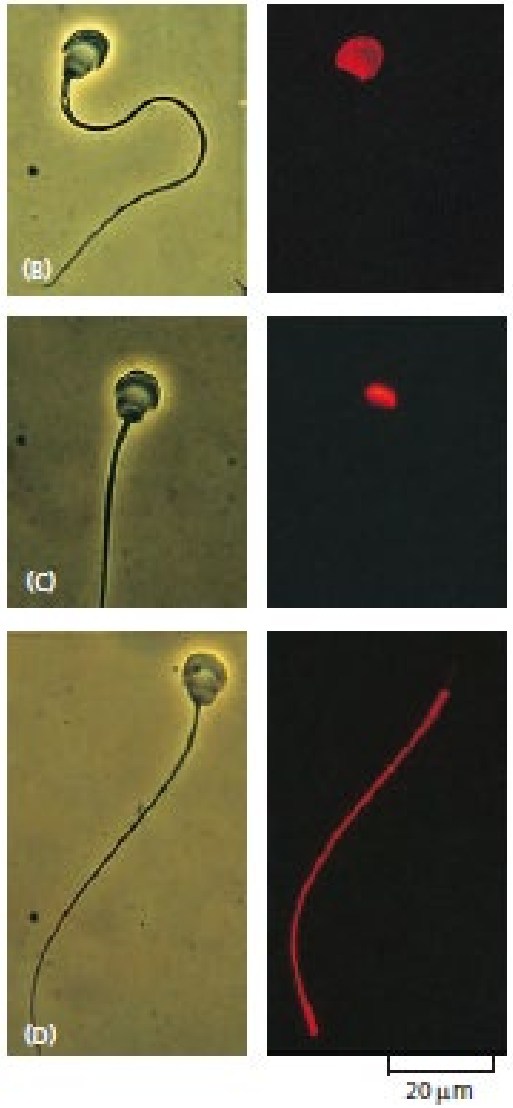
Measuring the rate of lateral diffusion of a membrane protein by fluorescence recovery after photobleaching



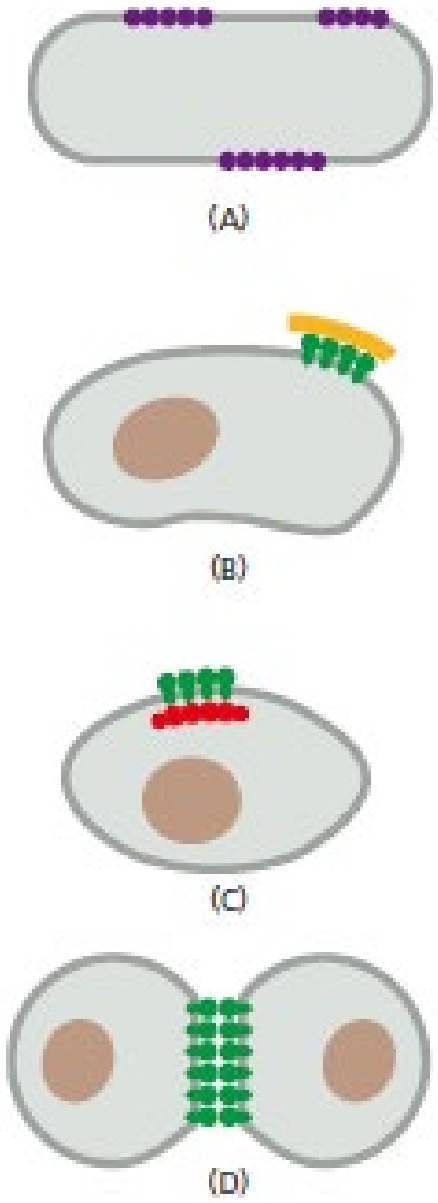
Cells Can Confine Proteins and Lipids to Specific Domains Within a Membrane







Four ways of restricting the lateral mobility of specific plasma membrane proteins

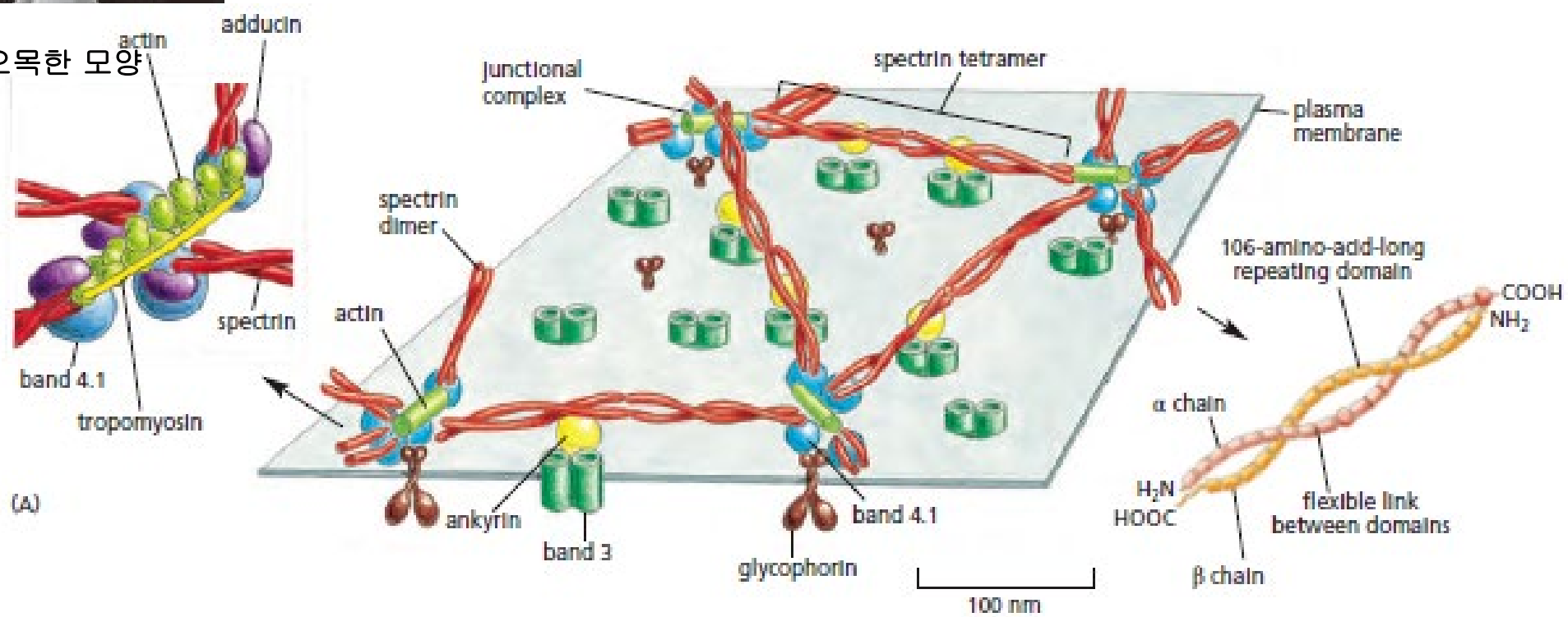


(A) 단백질은 큰 응집체로 자가 조립
(B) 세포 외부의 거대 분자 어셈블리와의 상호 작용
(C) 세포 내부의 거대 분자 어셈블리와의 상호 작용
(D) 다른 세포 표면의 단백질과 상호작용

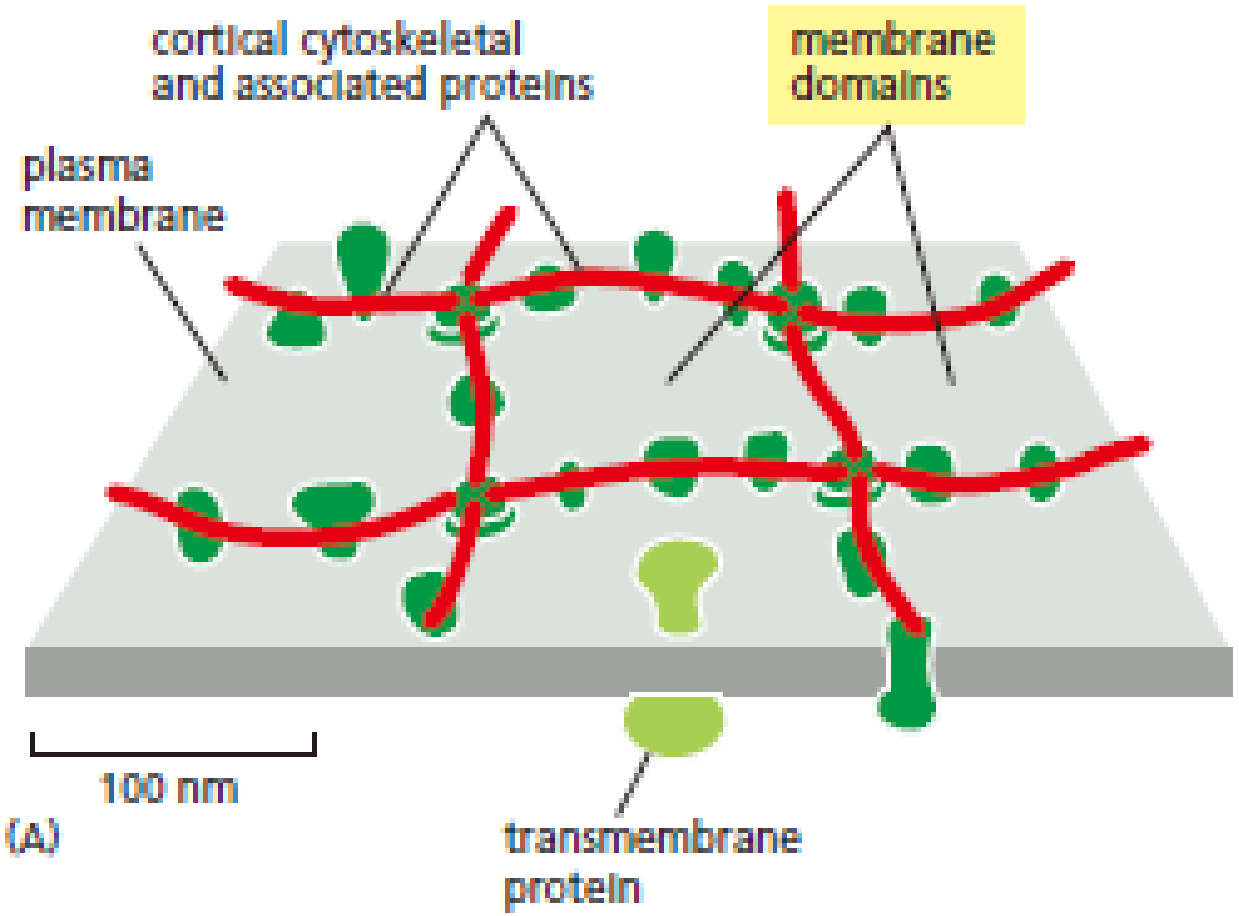
The Cortical Cytoskeleton Gives Membranes Mechanical Strength and Restricts Membrane Protein Diffusion

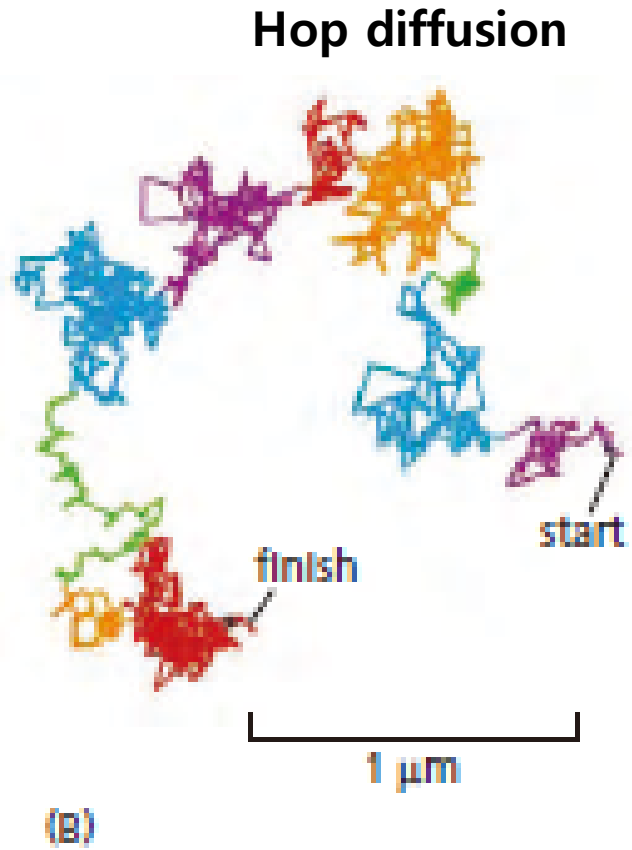
적혈구의 특징적인 양면 오목한 모양

Fence and Picket model

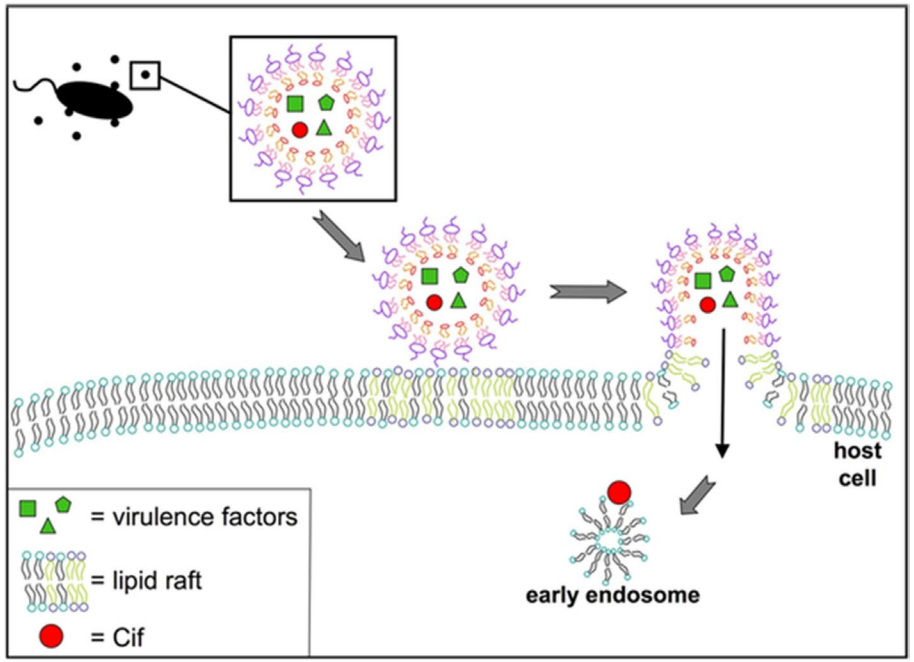


Corralling plasma membrane proteins by cortical cytoskeletal filaments

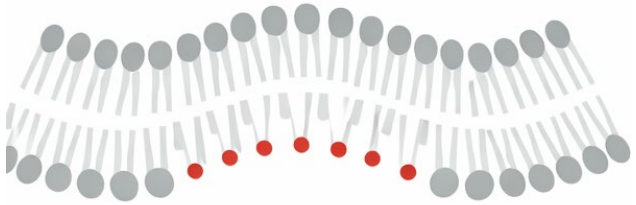


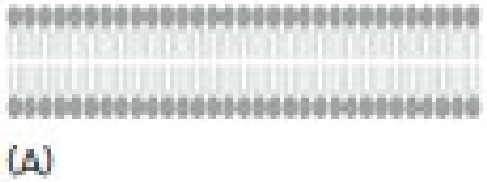


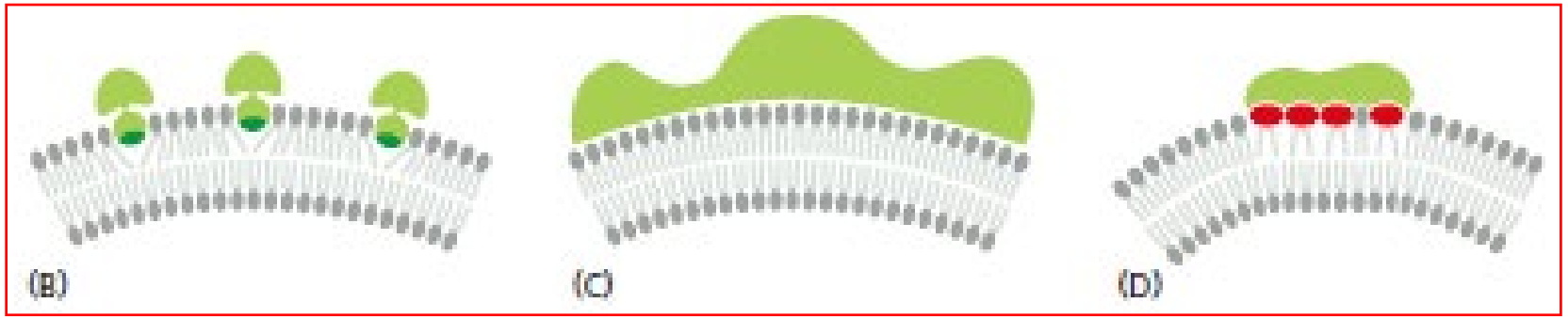
Membrane-bending Proteins Deform Bilayers



https://doi.org/10.3389/fmicb.2020.00432







parky5@ajou.ac.kr

감사합니다.